CHAPTER 1

Determinant lines and determinant line bundles

This appendix is an exposition of G. Segal's work sketched in [**?**] on determinant line bundles over the moduli spaces of Riemann surfaces with parametrized boundaries, especially in the genus-zero case. It requires some basic knowledge of complex analysis, functional analysis, differential geometry and geometric analysis, and is written in a self-contained way assuming this knowledge. We supply in this appendix all the necessary definitions, results and proofs needed in this work.

After the present work, including a version of this appendix, was finished, the author obtained a copy of [**?**] from D. Freed and a copy of [**?**] from P. Deligne. The second version [**?**] of Segal's manuscript on conformal field theory includes an appendix on determinant lines and determinant line bundles and contains more details than [**?**]. But the methods used in this appendix are different from those in [**?**] and many details omitted in [**?**] and [**?**] are given in the present appendix. Though a large part of the results and proofs in Sections D.3–D.6 follow the ideas in [**?**], the proofs and even the statements of some results (for example, Theorem 1.4.4 and Proposition 1.4.8) are given, to our knowledge, for the first time in the present appendix, which can be read as a separate paper on determinant lines and determinant line bundles.

The letter [**?**] contains a discussion of one-dimensional modular functors and emphasizes that determinant lines are $\mathbb{Z}_2$-graded lines and that there are some subtle compatibility conditions. We incorporate this point of view into our treatment, but we also note that in the case in which we are mostly interested in this monograph, that is, the case of genus-zero surfaces with at least one outgoing (positively oriented) boundary component, the issues related to the $\mathbb{Z}_2$-gradings of the determinant lines in fact become trivial (see Remark **??** and Section D.6). Also, this appendix incorporates answers to a number of questions raised by Deligne and an alternative argument in the proof of Proposition 1.4.8 supplied by him.

Section D.1 is a summary of the definitions and some properties of certain classes of linear operators on Hilbert spaces. Section D.2

is a discussion on determinant lines of Fredholm operators. In Section D.3 we define the determinant line over a Riemann surface with parametrized boundary and the determinant line bundles over the moduli spaces of Riemann surfaces with parametrized boundaries. In Section D.4 we construct the canonical isomorphisms between determinant lines corresponding to the sewing of Riemann surfaces with parametrized boundaries. We also construct the canonical connections on the determinant line bundles over moduli spaces of genus-zero Riemann surfaces with parametrized boundaries in Section D.4. In Section D.5, we show that these determinant lines together with the canonical isomorphisms constructed in Section D.4 give the basic central extension of the diffeomorphism group of S^1 in a natural way. In Section D.6 we give a definition of a variant of one-dimensional genus-zero modular functor and prove that any such one-dimensional genus-zero modular functor is a power of the determinant line bundle over the moduli space of genus-zero Riemann surfaces with parametrized boundaries.

1.1. Some classes of bounded linear operators

We give the definitions and list the basic properties of Fredholm, compact, finite rank, trace-class and Hilbert-Schmidt operators in this section. The reader can find the details in [**?**], [**?**] and [**?**]. For convenience, we shall always assume in this appendix that the Hilbert spaces we consider are separable and over $\mathbb{C}$. We shall always use I_V to denote the identity operator on a vector space V.

Definition 1.1.1. *Let H_1 and H_2 be Hilbert spaces. A bounded linear operator $F : H_1 \to H_2$ is called a Fredholm operator if both* $\mathrm{Ker}F$ *and* $\mathrm{Coker}F$ *are finite-dimensional. The index of F is the number*

$$\mathrm{Index}F = \dim \mathrm{Ker}F - \dim \mathrm{Coker}F.$$

A bounded linear operator $R : H_1 \to H_2$ is called a finite-rank operator if the image of R is finite-dimensional. A bounded linear operator $C : H_1 \to H_2$ is called a compact operator if it maps bounded subsets of H_1 to precompact subsets of H_2, that is, to subsets of H_2 whose closure is compact. A bounded linear operator $T : H_1 \to H_2$ is called a trace-class operator if there exists a sequence $\{\lambda_i\}_{i\in\mathbb{Z}_+}$ of complex numbers and orthonormal families $\{\phi_i\}_{i\in\mathbb{Z}_+}$ in H_1 and $\{\psi_i\}_{i\in\mathbb{Z}_+}$ in H_2 such that $\sum_{i\in\mathbb{Z}_+} |\lambda_i| < \infty$ and for any $h \in H_1$,

$$T(h) = \sum_{i\in\mathbb{Z}_+} \lambda_i(\phi_i, h)_1\psi_i,$$

where $(\cdot,\cdot)_1$ *is the inner product of* H_1*. The trace of* T *is the number*

$$\mathrm{tr}(T) = \sum_{i\in\mathbb{Z}_+} \lambda_i.$$

A bounded linear operator $T : H_1 \to H_2$ *is called a Hilbert-Schmidt operator if for any complete orthonormal basis* $\{e_i\}_{i\in\mathbb{Z}_+}$ *of* H_1*, the series* $\sum_{i\in\mathbb{Z}_+} \|Te_i\|^2$ *converges.*

Proposition 1.1.2. *The operators defined above have the following properties:*

(1) *Finite-rank operators are trace-class operators. Trace-class operators and Hilbert-Schmidt operators are compact operators.*
(2) *Let* C *be a compact operator (respectively, trace-class or Hilbert-Schmidt operator). For any bounded linear operators* $P_1 : H_0 \to H_1$ *and* $P_2 : H_2 \to H_3$*, the products* CP_1 *and* P_2C *are still compact operators (trace-class operators or Hilbert-Schmidt operators).*
(3) *The adjoint operator of a compact operator is also compact.*
(4) *The space of all compact operators from* H_1 *to* H_2 *is closed in the space of all bounded linear operators from* H_1 *to* H_2*. The space of compact operators are the closure of the space of finite-rank operators.*
(5) *The sum of a Fredholm operator* F *and a compact operator* C *(in particular, a finite rank operator or a trace-class operator) is a Fredholm operator. The index of* $F + C$ *is equal to the index of* C*.*
(6) *Products of Hilbert-Schmidt operators are trace-class operators.* ∎

Let $F : H_1 \to H_2$ be a Fredholm operator. A *quasi-inverse* (or *parametrix*) of F is a bounded linear operator $G : H_2 \to H_1$ such that GF is equal to the identity of H_1 plus a compact operator on H_1 and FG is equal to the identity of H_2 plus a compact operator on H_2. The following classical theorem is due to Atkinson whose proof can be found in the references listed in the beginning of this section:

Theorem 1.1.3 (F. W. Atkinson). *A bounded linear operator* $F : H_1 \to H_2$ *is Fredholm if and only if it has a quasi-inverse.* ∎

Remark 1.1.4. *From Theorem 1.1.3, we see that any quasi-inverse* G *of a Fredholm operator* F *is also a Fredholm operator.*

Remark 1.1.5. *In* [**?**]*, a canonical quasi-inverse* G *of* F *was constructed explicitly for each choice of closed subspaces* X_0 *and* Y_0 *of* H_1

and H_2, respectively, such that

$$H_1 = X_0 \oplus \mathrm{Ker}F$$

and

$$H_2 = Y_0 \oplus \mathrm{Im}F.$$

This quasi-inverse has the additional property that $FG - I_{H_2}$ and $GF - I_{H_1}$ are finite-rank operators. Since in our case both H_1 and H_2 are Hilbert spaces, we shall canonically choose X_0 and Y_0 to be the orthogonal complements of $\mathrm{Ker}F$ and $\mathrm{Im}F$, respectively, in H_1 and H_2. Thus we obtain a canonical quasi-inverse G of F such that $FG - I_{H_2}$ and $GF - I_{H_1}$ are finite-rank operators.

Let H be a Hilbert space. We denote the space of all trace-class operators on H by $\mathcal{I}_1$ and we define a norm $\|\cdot\|_1$ on $\mathcal{I}_1$ as follows: For any

$$T = \sum_{i\in\mathbb{Z}_+} \lambda_i(\phi_i, \cdot)\psi_i \in \mathcal{I}_1,$$
$$\|T\|_1 = \sum_{i\in\mathbb{Z}_+} |\lambda_i|.$$

Then $\mathcal{I}_1$ equipped with this norm is a Banach space. The proof of the following result due to Simon can be found in [**?**]:

THEOREM 1.1.6 (B. Simon). *There is a unique continuous function* $\det : I_H + \mathcal{I}_1 \to \mathbb{C}$ *with the following properties:*

(1) *For a complex variable z and $T \in \mathcal{I}_1$, $\det(I_H + zT)$ is an entire function satisfying*

$$|\det(I_H + zT)| \le e^{|z|\|T\|_1}.$$

(2) *For any $T_1, T_2 \in \mathcal{I}_1$,*

$$\det(I_H + T_1 + T_2 + T_1T_2) = \det(I_H + T_1)\det(I_H + T_2).$$

(3) *For any $T \in \mathcal{I}_1$, $\det(I_H + T) \neq 0$ if and only if $I_H + T$ is invertible. In this case, $(I_H + T)^{-1} - I_H$ is of trace class and*

$$\det(I_H + T)^{-1} = (\det(I_H + T))^{-1}.$$

(4) *Let T be a trace-class operator and $\lambda_n(T)$, $n \in \mathbb{Z}_+$, all the eigenvalues of T counting multiplicities. Then*

$$\det(I_H + T) = \prod_{n\in\mathbb{Z}_+} (1 + \lambda_n(T)).$$

∎

When a bounded linear operator on a Hilbert space H is of the form $I_H + T$ where T is of trace class, we say that the operator *has a determinant*. If H is finite-dimensional, any operator F on H has a determinant in the sense of the definition above and its determinant is equal to the determinant of an operator on a finite-dimensional space in the usual sense.

1.2. Determinant lines

In this section, we develop the theory of determinant lines of Fredholm operators. Many of the results are parallel to those for indices of Fredholm operators (see, for example, [**?**]). But for determinant lines, we need to construct canonical isomorphisms; proving the existence of isomorphisms is not enough. Thus many results here are harder and more subtle than the corresponding results for indices.

A $\mathbb{Z}_2$*-graded line* or simply *graded line* is a one-dimensional vector space with an element of $\mathbb{Z}_2$ called the *degree*. The degree of a graded line L is denoted by $\deg(L)$. Given any element $v \in L$, the *degree of* v is defined to be the degree of L and is denoted by $\deg(v)$. *Isomorphisms of graded lines* are defined in the obvious way. For an n-dimensional vector space V, we define the *determinant line* $\mathrm{Det}V$ *of* V to be the graded line $\bigwedge^n V$ of the highest exterior power of V with degree n modulo $2\mathbb{Z}$. It is clear that we can think of Det as a functor from the category of finite-dimensional vector spaces with isomorphisms as morphisms to the category of graded lines with isomorphisms as morphisms. Note our choice of notations in this appendix: Det will always denote a determinant line of a finite-dimensional space, a Fredholm operator or a Riemann surface with analytically parametrized boundary, while det will always denote the determinant (not the determinant line) of an operator having a determinant.

Given two vector spaces of the same finite dimension, they are always isomorphic, but there is no canonical isomorphism. When the two vector spaces are the images of an object in a category under two naturally isomorphic functors, the spaces are *canonically isomorphic*. In this appendix, we shall write canonical isomorphisms as $\simeq$. Note that when we say that two vector spaces are canonically isomorphic, we have already chosen a natural isomorphism understood from the context. The same conventions also apply to graded lines and so on.

The category of graded lines has a tensor product operation: Given any two graded lines L_1 and L_2, the one-dimensional vector space $L_1 \otimes L_2$ together with the degree

$$\deg(L_1 \otimes L_2) = \deg(L_1) + \deg(L_2)$$

is a graded line. This tensor product is a functor from the category of ordered pairs of graded lines to the category of graded lines. There is a natural isomorphism from this functor to itself called the *symmetry of graded lines* defined as follows: Let L_1 and L_2 be two graded lines. For any $v_1 \in L_1$ and $v_2 \in L_2$, we define an isomorphism from $L_1 \otimes L_2$ to $L_2 \otimes L_1$ by $v_1 \otimes v_2 \mapsto (-1)^{\deg(v_1)\deg(v_2)} v_2 \otimes v_2$. Clearly these isomorphisms give a natural isomorphism. (In fact, graded lines with the tensor product and this symmetry form a symmetric tensor category.) Thus $L_1 \otimes L_2 \simeq L_2 \otimes L_1$.

In this appendix, when we discuss canonical isomorphisms, we shall not indicate explicitly what the functors and the natural isomorphisms are, except in the proof of the following basic lemma:

LEMMA 1.2.1. *Let U be a finite-dimensional vector space and V a subspace of U. Then*

$$\mathrm{Det}U \simeq \mathrm{Det}(U/V) \otimes \mathrm{Det}V.$$

PROOF. We view $(U, V) \mapsto \mathrm{Det}U$ and $(U, V) \mapsto \mathrm{Det}(U/V) \otimes \mathrm{Det}V$ as functors from the category of pairs of finite-dimensional vector spaces and their subspaces to the category of graded lines. To show that

$$\mathrm{Det}U \simeq \mathrm{Det}(U/V) \otimes \mathrm{Det}V$$

is to construct a natural isomorphism between these two functors.

Assume that the dimensions of U and V are m and n, respectively. Then $n \leq m$. Let $v_1 \wedge \cdots \wedge v_n$ be a nonzero element of $\mathrm{Det}V$. Any nonzero element of $\mathrm{Det}U$ can be written as

$$u_1 \wedge \cdots \wedge u_{n-m} \wedge v_1 \wedge \cdots \wedge v_m$$

where $u_1, \ldots, u_{m-n}$ are elements of U not in V. Thus we have an element

$$((u_1 + V) \wedge \cdots \wedge (u_{n-m} + V)) \otimes v_1 \wedge \cdots \wedge v_n \in \mathrm{Det}U/V \otimes \mathrm{Det}V.$$

Whenever there are elements $\tilde{u}_1, \ldots, \tilde{u}_{m-n} \in U$ and $\tilde{v}_1, \ldots, \tilde{v}_n \in V$ such that

$$\tilde{u}_1 \wedge \cdots \wedge \tilde{u}_{n-m} \wedge \tilde{v}_1 \wedge \cdots \wedge \tilde{v}_m = u_1 \wedge \cdots \wedge u_{n-m} \wedge v_1 \wedge \cdots \wedge v_m,$$

we have $\tilde{u}_i = \sum_{k=1}^{m-n} a_{ik} u_k + \hat{v}_i$, $i = 1, \ldots, m-n$, and $\tilde{v}_j = \sum_{l=1}^{n} b_{jl} v_l$ where $\hat{v}_i$ are elements of V and (a_{ik}) and (b_{jl}) are matrices such that $\det(a_{ik}) \det(b_{jl}) = (\delta_{ij})$. Thus

$$\begin{aligned}&((\tilde{u}_1 + V) \wedge \cdots \wedge (\tilde{u}_{n-m} + V)) \otimes \tilde{v}_1 \wedge \cdots \wedge \tilde{v}_n \\ &\quad = ((u_1 + V) \wedge \cdots \wedge (u_{n-m} + V)) \otimes v_1 \wedge \cdots \wedge v_n,\end{aligned}$$

and we obtain a linear map from $\mathrm{Det}U$ to $\mathrm{Det}U/V \otimes \mathrm{Det}V$. It is clear that this map is an isomorphism and in fact natural. $\square$

Let H_1 and H_2 be Hilbert spaces and $F : H_1 \to H_2$ a Fredholm operator. Following Quillen [**?**], we associate to F a one-dimensional vector space

$$\mathrm{Det}F = \mathrm{Det}(\mathrm{Ker}F)^* \otimes \mathrm{Det}(\mathrm{Coker}F)$$

where on the right-hand side Det is the functor discussed above. There is a natural $\mathbb{Z}_2$-grading of $\mathrm{Det}F$ given by the index of F modulo $2\mathbb{Z}$. With this grading, $\mathrm{Det}F$ becomes a graded line. This graded line $\mathrm{Det}F$ is called the *determinant line of the Fredholm operator* F.

Let $F_1 : H_1 \to H_2$ and $F_2 : H_3 \to H_4$ be Fredholm operators. An *equivalence from* F_1 *to* F_2 is a pair $f_1 : H_1 \to H_3$ and $f_2 : H_2 \to H_4$ of bounded linear isomorphisms such that the diagram

$$\begin{tikzcd} H_1 \arrow[r, "F_1"] \arrow[d, "f_1"'] & H_2 \arrow[d, "f_2"] \\ H_3 \arrow[r, "F_2"'] & H_4 \end{tikzcd}$$

commutes. It is clear that the restriction of f_1 to $\mathrm{Ker}F_1$ gives a linear isomorphism from $\mathrm{Ker}F_1$ to $\mathrm{Ker}F_2$ and f_2 induces a linear isomorphism from $\mathrm{Coker}F_1$ to $\mathrm{Coker}F_2$. Thus the pair (f_1, f_2) induces a linear isomorphism

$$\mathrm{Det}_{f_1,f_2} : \mathrm{Det}F_1 \to \mathrm{Det}F_2.$$

Thus Det is a functor from the category of Fredholm operators with equivalences as morphisms to the category of one-dimensional vector spaces with isomorphisms as morphisms.

Note that linear operators between finite-dimensional spaces are Fredholm operators. Thus they have determinant lines.

Proposition 1.2.2. *Let H_1 and H_2 be finite-dimensional vector spaces and $F : H_1 \to H_2$ a linear operator. Then there exists a canonical isomorphism from* $\mathrm{Det}F$ *to* $(\mathrm{Det}H_1)^* \otimes \mathrm{Det}H_2$ *such that in the case that $H_1 = H_2 = H$ and F is invertible, the canonical isomorphism from* $\mathbb{C} = \mathrm{Det}F$ *to* $\mathbb{C} \simeq (\mathrm{Det}H)^* \otimes \mathrm{Det}H$ *is given by multiplying* $(\det F)^{-1} = \det F^{-1}$ *to elements of* $\mathbb{C}$.

Proof. Since $H_1/\mathrm{Ker}F \simeq \mathrm{Im}F$, we have

$$\mathrm{Det}(H_1/\mathrm{Ker}F) \simeq \mathrm{Det}\mathrm{Im}F,$$

or equivalently,

$$\mathrm{Det}(H_1/\mathrm{Ker}F) \otimes (\mathrm{Det}\mathrm{Im}F)^* \simeq \mathbb{C}.$$

By Lemma 1.2.1, we have

$$\mathrm{Det}(H_1/\mathrm{Ker}F) \simeq \mathrm{Det}H_1 \otimes (\mathrm{Det}\mathrm{Ker}F)^*$$

and

$$(\mathrm{Det}\mathrm{Im}F)^* \simeq \mathrm{Det}H_2 \otimes \mathrm{Det}\mathrm{Coker}F.$$

Thus

$$\begin{aligned}\mathbb{C} \simeq & \mathrm{Det}(H_1/\mathrm{Ker}F) \otimes (\mathrm{Det}\mathrm{Im}F)^* \\ \simeq & \mathrm{Det}H_1 \otimes (\mathrm{Det}\mathrm{Ker}F)^* \otimes (\mathrm{Det}H_2)^* \otimes \mathrm{Det}\mathrm{Coker}F \\ \simeq & \mathrm{Det}H_1 \otimes (\mathrm{Det}H_2)^* \otimes \mathrm{Det}F,\end{aligned}$$

or equivalently, $\mathrm{Det}F \simeq (\mathrm{Det}H_1)^* \otimes \mathrm{Det}H_2$.

In the case that $H_1 = H_2 = H$ and F is invertible, we have $\mathrm{Ker}F = 0$ and $\mathrm{Im}F = H$. Thus the canonical isomorphism from $H_1/\mathrm{Ker}F = H$ to $\mathrm{Im}F = H$ is F itself and the canonical isomorphism from $\mathrm{Det}(H_1/\mathrm{Ker}F) = \mathrm{Det}H$ to $\mathrm{Det}F = \mathrm{Det}H$ is given by multiplying $\det F$ to elements of $\mathrm{Det}H$. From the construction above, we see that the canonical isomorphism from $\mathbb{C} = \mathrm{Det}F$ to $\mathbb{C} \simeq (\mathrm{Det}H)^* \otimes \mathrm{Det}H$ is indeed given by multiplying $(\det F)^{-1} = \det F^{-1}$ to elements of $\mathbb{C}$. $\square$

Proposition 1.2.3 (Snake lemma). *Consider the following diagram:*

$$\begin{tikzcd} 0 \arrow[r] & H_1 \arrow[r] \arrow[d, "F_1"'] & H_2 \arrow[r] \arrow[d, "F_2"'] & H_3 \arrow[r] \arrow[d, "F_3"'] & 0 \\ 0 \arrow[r] & H_4 \arrow[r] & H_5 \arrow[r] & H_6 \arrow[r] & 0 \end{tikzcd} \tag{1}$$

where H_i, $i = 1, \ldots, 6$, are Hilbert spaces, F_1, F_2, F_3 are bounded linear operator, and the horizontal lines are exact. If two of the operators F_1, F_2, F_3 are Fredholm, then the other is also Fredholm and there is a canonical isomorphism from $\mathrm{Det}F_1 \otimes \mathrm{Det}F_3$ *to* $\mathrm{Det}F_2$.

Proof. The diagram (1) gives an exact sequence

$$\begin{aligned} 0 \longrightarrow \mathrm{Ker}F_1 \longrightarrow \mathrm{Ker}F_2 \xrightarrow{i} \mathrm{Ker}F_3 \xrightarrow{k} \\ \xrightarrow{k} \mathrm{Coker}F_1 \xrightarrow{j} \mathrm{Coker}F_2 \longrightarrow \mathrm{Coker}F_3 \longrightarrow 0. \end{aligned} \tag{2}$$

Assume that F_1 and F_3 are Fredholm. From the exact sequence (2), we obtain

$$\begin{aligned} \mathrm{Im}i &\simeq \mathrm{Ker}F_2/\mathrm{Ker}i, \\ \mathrm{Ker}i &\simeq \mathrm{Ker}F_1, \\ \mathrm{Im}i &\subset \mathrm{Ker}F_3. \end{aligned} \tag{3}$$

Since F_1 and F_3 are Fredholm, both $\mathrm{Ker}F_1$ and $\mathrm{Ker}F_3$ are finite-dimensional. Thus by (**??**)—(3), $\mathrm{Ker}F_2$ is also finite-dimensional. Similarly, we can

show that $\mathrm{Coker}F_2$ is finite-dimensional and consequently F_2 is Fredholm. If F_1, F_2 or F_2, F_3 are Fredholm, we can show similarly that F_3 or F_1, respectively, is Fredholm.

The dual of (2)

$$\begin{array}{ccccccc} 0 & \longleftarrow & (\mathrm{Ker}F_1)^* & \longleftarrow & (\mathrm{Ker}F_2)^* & \xleftarrow{i^*} & \\ & \xleftarrow{i^*} & (\mathrm{Ker}F_3)^* & \xleftarrow{k^*} & (\mathrm{Coker}F_1)^* & \xleftarrow{j^*} & \\ & \xleftarrow{j^*} & (\mathrm{Coker}F_2)^* & \longleftarrow & (\mathrm{Coker}F_3)^* & \longleftarrow & 0 \end{array} \tag{4}$$

is also exact. From the exact sequences (2) and (4), we see that

$$(\mathrm{Ker}F_1)^* \simeq (\mathrm{Ker}F_2)^*/\mathrm{Im}i^*, \tag{5}$$

$$\mathrm{Coker}F_3 \simeq \mathrm{Coker}F_2/\mathrm{Im}j, \tag{6}$$

$$\mathrm{Im}i^* \simeq (\mathrm{Ker}F_3)^*/\mathrm{Im}k^*, \tag{7}$$

$$\mathrm{Im}j \simeq \mathrm{Coker}F_1/\mathrm{Im}k. \tag{8}$$

By Lemma 1.2.1 and (5)–(8), we have

$$\begin{aligned} &\mathrm{Det}F_1 \otimes \mathrm{Det}F_3 \\ &\quad = \mathrm{Det}(\mathrm{Ker}F_1)^* \otimes \mathrm{DetCoker}F_1 \otimes \mathrm{Det}(\mathrm{Ker}F_3)^* \otimes \mathrm{DetCoker}F_3 \\ &\quad \simeq \mathrm{Det}((\mathrm{Ker}F_2)^*/\mathrm{Im}i^*) \otimes \mathrm{DetCoker}F_1 \otimes \\ &\qquad \otimes \mathrm{Det}(\mathrm{Ker}F_3)^* \otimes \mathrm{Det}(\mathrm{Coker}F_2/\mathrm{Im}j) \\ &\quad \simeq \mathrm{Det}((\mathrm{Ker}F_2)^*/\mathrm{Im}i^*) \otimes (\mathrm{DetIm}j \otimes \mathrm{DetIm}k) \otimes \\ &\qquad \otimes (\mathrm{DetIm}i^* \otimes \mathrm{DetIm}k^*) \otimes \mathrm{Det}(\mathrm{Coker}F_2/\mathrm{Im}j) \\ &\quad \simeq \mathrm{Det}((\mathrm{Ker}F_2)^*/\mathrm{Im}i^*) \otimes \mathrm{DetIm}i^* \otimes \\ &\qquad \otimes \mathrm{Det}(\mathrm{Coker}F_2/\mathrm{Im}j) \otimes \mathrm{DetIm}j \otimes \mathrm{DetIm}k \otimes \mathrm{DetIm}k^* \\ &\quad \simeq \mathrm{Det}(\mathrm{Ker}F_2)^* \otimes \mathrm{DetCoker}F_2 \otimes \mathrm{DetIm}k \otimes \mathrm{DetIm}k^* \\ &\quad \simeq \mathrm{Det}F_2 \otimes \mathrm{DetIm}k \otimes \mathrm{DetIm}k^*. \end{aligned} \tag{9}$$

It is easy to see that $\mathrm{Im}k^*$ is in fact canonically isomorphic to $(\mathrm{Im}k)^*$. The canonical paring between $\mathrm{Im}k$ and $(\mathrm{Im}k)^*$ induces a nondegenerate pairing between $\mathrm{DetIm}k$ and $\mathrm{Det}(\mathrm{Im}k)^*$ which can be regarded as a canonical isomorphism from $\mathrm{DetIm}k \otimes \mathrm{Det}(\mathrm{Im}k)^*$ to $\mathbb{C}$. Thus

$$\mathrm{DetIm}k \otimes \mathrm{DetIm}k^* \simeq \mathbb{C}. \tag{10}$$

The second conclusion follows from (9) and (10). □

Corollary 1.2.4. *Let $F_1 : H_1 \to H_2$ and $F_2 : H_2 \to H_3$ be Fredholm operators. Then* $\mathrm{Det}(F_2F_1)$ *is canonically isomorphic to* $\mathrm{Det}F_2 \otimes \mathrm{Det}F_1$.

PROOF. Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H_1 & \xrightarrow{i} & H_1 \oplus H_2 & \xrightarrow{p} & H_2 & \longrightarrow & 0 \\ & & {\scriptstyle F_1}\big\downarrow & & {\scriptstyle F_2F_1 \oplus I_{H_2}}\big\downarrow & & {\scriptstyle F_2}\big\downarrow & & \\ 0 & \longrightarrow & H_2 & \xrightarrow{j} & H_3 \oplus H_2 & \xrightarrow{q} & H_3 & \longrightarrow & 0 \end{array}$$

where i, j, p, q are defined by

$$\begin{aligned} i(u) &= (u, F_1 u), \\ j(v) &= (F_2 v, v), \\ p(u, v) &= F_1(u) - v, \\ q(w, v) &= w - F_2(v) \end{aligned}$$

for all $u \in H_1, v \in H_2$ and $w \in H_3$. This diagram is commutative and the horizontal sequences are exact. Applying Proposition 1.2.3, we see that $\mathrm{Det}(F_2F_1 \oplus I_{H_2})$ is canonically isomorphic to $\mathrm{Det}F_2 \otimes \mathrm{Det}F_2$. Since $\mathrm{Det}(F_2F_1)$ is canonically isomorphic to $\mathrm{Det}(F_2F_1 \oplus I_{H_2})$, it is also canonically isomorphic to $\mathrm{Det}F_2 \otimes \mathrm{Det}F_2$. □

COROLLARY 1.2.5. *Let $R : H \to H$ be a finite-rank operator. Then there is a canonical isomorphism from* $\mathrm{Det}(I_H + R)$ *to* $\mathbb{C}$ *such that in the case that $I_H + R$ is invertible, this canonical isomorphism from* $\mathbb{C} = \mathrm{Det}(I_H + R)$ *to* $\mathbb{C}$ *is given by multiplying the determinant* $\det(I_H + R)^{-1}$ *to elements of* $\mathbb{C}$.

PROOF. Let h be any finite-dimensional vector space containing $\mathrm{Im}R$ and $(I_H + R)_h$ (the restriction of $I_H + R$ to h). Consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & h & \longrightarrow & H & \longrightarrow & H/h & \longrightarrow & 0 \\ & & {\scriptstyle (I_H+R)_h}\big\downarrow & & {\scriptstyle I_H+R}\big\downarrow & & {\scriptstyle I_{H/h}}\big\downarrow & & \\ 0 & \longrightarrow & h & \longrightarrow & H & \longrightarrow & H/h & \longrightarrow & 0. \end{array}$$

The commutativity of the diagram and the exactness of the rows is clear. By Proposition 1.2.3, $\mathrm{Det}(I_H + R)$ is canonically isomorphic to $\mathrm{Det}(I_H + R)_h \otimes \mathrm{Det}I_{H/h}$. Since h is finite-dimensional, $(I_H + R)_h$ is an operator on a finite-dimensional space. By Proposition 1.2.2, $\mathrm{Det}(I_H + R)_h \simeq \mathbb{C}$. Since $\mathrm{Det}I_{H/h} = \mathbb{C}$, we obtain a canonical isomorphism from $\mathrm{Det}(I_H + R)$ to $\mathbb{C}$ by the snake lemma.

In the case that $I_H + R$ is invertible, by the construction above and Proposition 1.2.2, the canonical isomorphism from $\mathbb{C} = \mathrm{Det}(I_H + R)_h$ to $\mathbb{C}$ is given by multiplying the determinant $\det(I_H + R)_h^{-1}$ to elements of $\mathbb{C}$. Let λ_n, $n = 1, \ldots, N$, be the eigenvalues of R. Since R is of finite rank, N must be finite. Since h contains $\mathrm{Im}R$, λ_n, $n = 1, \ldots, N$, are

also the eigenvalues of R_h (the restriction of R to h). Thus by Theorem 1.1.6

$$\begin{aligned}\det(I_H+R)_h^{-1} &= (\det(I_h+R_h))^{-1}\\ &= \left(\prod_{n=1}^{N}(1+\lambda_n)\right)^{-1}\\ &= (\det(I_H+R))^{-1}\\ &= \det(I_H+R)^{-1},\end{aligned}$$

proving the corollary. □

We also have the following associativity:

Proposition 1.2.6. *Let $F_1 : H_1 \to H_2$, $F_2 : H_2 \to H_3$, $F_3 : H_3 \to H_4$ be Fredholm operators. Then the diagram*

$$\begin{array}{ccc} \mathrm{Det}(F_3F_2F_1) & \longrightarrow & \mathrm{Det}F_3 \otimes \mathrm{Det}(F_2F_1) \\ \downarrow & & \downarrow \\ \mathrm{Det}(F_3F_2) \otimes \mathrm{Det}F_1 & \longrightarrow & \mathrm{Det}F_3 \otimes \mathrm{Det}F_2 \otimes \mathrm{Det}F_1 \end{array} \tag{11}$$

obtained using the canonical isomorphisms constructed in Corollary 1.2.4 is commutative.

Proof. We prove this result by writing down the canonical isomorphisms in the diagram (11) explicitly. First we consider the canonical isomorphism from $\mathrm{Det}(F_2F_1)$ to $\mathrm{Det}F_2 \otimes \mathrm{Det}F_1$ which induces the right vertical arrow in (11). In fact, the construction below is a direct proof of Corollary 1.2.4. As in the proof of the snake lemma, we have the following exact sequence:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Ker}F_1 & \longrightarrow & \mathrm{Ker}(F_2F_1 \oplus I_{H_2}) & \longrightarrow & \\ & \longrightarrow & \mathrm{Ker}F_2 & \longrightarrow & \mathrm{Coker}F_1 & \longrightarrow & \\ & \longrightarrow & \mathrm{Coker}(F_2F_1 \oplus I_{H_2}) & \longrightarrow & \mathrm{Coker}F_2 & \longrightarrow & 0. \end{array}$$

Since $\mathrm{Ker}(F_2F_1 \oplus I_{H_2}) \simeq \mathrm{Ker}(F_2F_1)$ and $\mathrm{Coker}(F_2F_1 \oplus I_{H_2}) \simeq \mathrm{Coker}(F_2F_1)$, this exact sequence gives us the exact sequence

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Ker}F_1 & \longrightarrow & \mathrm{Ker}(F_2F_1) & \xrightarrow{i_1} & \mathrm{Ker}F_2 & \xrightarrow{k_1} & \\ & \xrightarrow{k_1} & \mathrm{Coker}F_1 & \xrightarrow{j_1} & \mathrm{Coker}(F_2F_1) & \longrightarrow & \mathrm{Coker}F_2 & \longrightarrow & 0 \end{array}$$

and its dual

$$\begin{array}{ccccccc} 0 & \longleftarrow & (\mathrm{Ker}F_1)^* & \longleftarrow & (\mathrm{Ker}(F_2F_1))^* & \xleftarrow{i_1^*} & \\ & \xleftarrow{i_1^*} & (\mathrm{Ker}F_2)^* & \xleftarrow{k_1^*} & (\mathrm{Coker}F_1)^* & \xleftarrow{j_1^*} & \\ & \xleftarrow{j_1^*} & (\mathrm{Coker}(F_2F_1))^* & \longleftarrow & (\mathrm{Coker}F_2)^* & \longleftarrow & 0. \end{array}$$

Thus similar to (5)–(8), we have

$$\begin{aligned} (\mathrm{Ker}F_1)^* &\simeq (\mathrm{Ker}(F_2F_1))^*/\mathrm{Im}i_1^*, \\ \mathrm{Coker}F_2 &\simeq \mathrm{Coker}(F_2F_1)/\mathrm{Im}j_1, \\ \mathrm{Im}i_1^* &\simeq (\mathrm{Ker}F_2)^*/\mathrm{Im}k_1^*, \\ \mathrm{Im}j_1 &\simeq \mathrm{Coker}F_1/\mathrm{Im}k_1. \end{aligned}$$

From these formulas and the fact that $\mathrm{Det}\mathrm{Im}k_1^* \simeq (\mathrm{Det}\mathrm{Im}k_1)^*$, we have

$$\mathrm{Det}(\mathrm{Ker}F_1)^* \otimes \mathrm{Det}\mathrm{Im}i_1^* \simeq \mathrm{Det}(\mathrm{Ker}(F_2F_1))^*, \tag{12}$$

$$\mathrm{Det}\mathrm{Coker}F_2 \otimes \mathrm{Det}\mathrm{Im}j_1 \simeq \mathrm{Det}\mathrm{Coker}(F_2F_1), \tag{13}$$

$$\mathrm{Det}\mathrm{Im}i_1^* \otimes (\mathrm{Det}\mathrm{Im}k_1)^* \simeq \mathrm{Det}(\mathrm{Ker}F_2)^*, \tag{14}$$

$$\mathrm{Det}\mathrm{Im}j_1 \otimes \mathrm{Det}\mathrm{Im}k_1 \simeq \mathrm{Det}\mathrm{Coker}F_1. \tag{15}$$

Let u_1^*, u_2^*, u_3, u_4, u_5 be basis (nonzero vectors) of $\mathrm{Det}(\mathrm{Ker}F_1)^*$, $\mathrm{Det}\mathrm{Im}i_1^*$, $\mathrm{Det}\mathrm{Coker}F_2$, $\mathrm{Det}\mathrm{Im}j_1$, $\mathrm{Det}\mathrm{Im}k_1$, respectively, and $u_5^* \in (\mathrm{Det}\mathrm{Im}k_1)^*$ the dual basis of u_5. We identify the right-hand sides of (12)–(15) with their left-hand sides, respectively. Then under these identifications, $u_1^* \otimes u_2^*$, $u_3 \otimes u_4$, $u_2^* \otimes u_5^*$ and $u_4 \otimes u_5$ are basis of $\mathrm{Det}(\mathrm{Ker}(F_2F_1))^*$, $\mathrm{Det}\mathrm{Coker}(F_2F_1)$, $\mathrm{Det}(\mathrm{Ker}F_2)^*$ and $\mathrm{Det}\mathrm{Coker}F_1$, respectively. Thus $u_1^* \otimes u_2^* \otimes u_3 \otimes u_4$ and $u_2^* \otimes u_5^* \otimes u_3 \otimes u_1^* \otimes u_4 \otimes u_5$ are basis of $\mathrm{Det}(F_2F_1)$ and $\mathrm{Det}F_2 \otimes \mathrm{Det}F_1$, respectively, and the canonical isomorphism from $\mathrm{Det}(F_2F_1)$ to $\mathrm{Det}F_2 \otimes \mathrm{Det}F_1$ is the unique linear map given by

$$u_1^* \otimes u_2^* \otimes u_3 \otimes u_4 \mapsto u_2^* \otimes u_5^* \otimes u_3 \otimes u_1^* \otimes u_4 \otimes u_5.$$

Repeat the above discussion with F_1 and F_2 replaced by F_2F_1 and F_3, respectively, and denote the maps i_1, j_1, k_1 in this case by i_2, j_2, k_2, respectively. Let u_6^*, u_7, u_8, u_9 be basis of $\mathrm{Det}\mathrm{Im}i_2^*$, $\mathrm{Det}\mathrm{Im}j_2$, $\mathrm{Det}\mathrm{Coker}F_3$, $\mathrm{Det}\mathrm{Im}k_2$, respectively, and $u_9^* \in (\mathrm{Det}\mathrm{Im}k_2)^*$ the dual basis of u_9. Then $u_1^* \otimes u_2^* \otimes u_6^*$, $u_7 \otimes u_8$, $u_6^* \otimes u_9^*$ are basis of $\mathrm{Det}(\mathrm{Ker}(F_3F_2F_1))^*$, $\mathrm{Det}\mathrm{Coker}(F_3F_2F_1)$, $\mathrm{Det}(\mathrm{Ker}f_3)^*$, respectively. Thus $u_1^* \otimes u_2^* \otimes u_6^* \otimes u_7 \otimes u_8$ and

$$u_6^* \otimes u_9^* \otimes u_8 \otimes u_1^* \otimes u_2^* \otimes u_3 \otimes u_4$$

are basis of $\mathrm{Det}(F_3F_2F_1)$ and $\mathrm{Det}F_3 \otimes \mathrm{Det}(F_2F_1)$, respectively. Note that we have

$$\mathrm{Det}\mathrm{Coker}F_2 \otimes \mathrm{Det}\mathrm{Im}j_1 \simeq \mathrm{Det}\mathrm{Coker}(F_2F_1) \simeq \mathrm{Det}\mathrm{Im}j_2 \otimes \mathrm{Det}\mathrm{Im}k_2$$

and the resulting canonical isomorphism from $\mathrm{Det}\mathrm{Coker}F_2 \otimes \mathrm{Det}\mathrm{Im}j_1$ to $\mathrm{Det}\mathrm{Im}j_2 \otimes \mathrm{Det}\mathrm{Im}k_2$ is determined by $u_3 \otimes u_4 \mapsto u_7 \otimes u_9$. Since we identify both $\mathrm{Det}\mathrm{Coker}F_2 \otimes \mathrm{Det}\mathrm{Im}j_1$ and $\mathrm{Det}\mathrm{Im}j_2 \otimes \mathrm{Det}\mathrm{Im}k_2$ with $\mathrm{Det}\mathrm{Coker}(F_2F_1)$, $u_3 \otimes u_4$ and $u_7 \otimes u_9$ are the same in $\mathrm{Det}\mathrm{Coker}(F_2F_1)$. Thus

$$u_6^* \otimes u_9^* \otimes u_8 \otimes u_1^* \otimes u_2^* \otimes u_3 \otimes u_4$$

and

$$u_6^* \otimes u_9^* \otimes u_8 \otimes u_1^* \otimes u_2^* \otimes u_7 \otimes u_9$$

are the same in $\mathrm{Det}F_3 \otimes \mathrm{Det}(F_2F_1)$. By construction, the canonical isomorphism from $\mathrm{Det}(F_3F_2F_1)$ to $\mathrm{Det}F_3 \otimes \mathrm{Det}(F_2F_1)$ is given by

$$u_1^* \otimes u_2^* \otimes u_6^* \otimes u_7 \otimes u_8 \mapsto u_6^* \otimes u_9^* \otimes u_8 \otimes u_1^* \otimes u_2^* \otimes u_7 \otimes u_9,$$

or equivalently, given by

$$u_1^* \otimes u_2^* \otimes u_6^* \otimes u_7 \otimes u_8 \mapsto u_6^* \otimes u_9^* \otimes u_8 \otimes u_1^* \otimes u_2^* \otimes u_3 \otimes u_4. \tag{16}$$

From the discussions above, we see that the upper horizontal arrow in (11) is given by (16) and the right vertical arrow in (11) is given by

$$u_6^* \otimes u_9^* \otimes u_8 \otimes u_1^* \otimes u_2^* \otimes u_3 \otimes u_4 \mapsto u_6^* \otimes u_9^* \otimes u_8 \otimes u_2^* \otimes u_5^* \otimes u_3 \otimes u_1^* \otimes u_4 \otimes u_5.$$

Thus the composition of the upper horizontal arrow and the right vertical arrow in (11) is given by

$$u_1^* \otimes u_2^* \otimes u_6^* \otimes u_7 \otimes u_8 \mapsto u_6^* \otimes u_9^* \otimes u_8 \otimes u_2^* \otimes u_5^* \otimes u_3 \otimes u_1^* \otimes u_4 \otimes u_5. \tag{17}$$

One can prove using the same method as above to show that the composition of the left vertical arrow and the lower horizontal arrow in (11) is also given by (17). Thus the diagram (11) is commutative. □

Remark 1.2.7. *Consider the following two categories: The objects of both categories are Hilbert spaces. The morphisms between two Hilbert spaces in the first category are the Fredholm operators from the domain to the codomain, while the morphisms in the second category are the determinant lines of the Fredholm operators from the domain to the codomain. Then the above proposition shows that we can interpret* $F \mapsto \mathrm{Det}F$ *as a functor from the first category to the second category.*

Remark 1.2.8. *In the discussions above, if we take all the spaces to be vector spaces and all the maps to be linear maps, the conclusions are still true. So the results above are all algebraic in nature. But we do need Hilbert space structures below to obtain deeper results.*

Proposition 1.2.9. *Let* $T : H \to H$ *be a trace-class operator. Then there exists a canonical isomorphism from* $\mathrm{Det}(I_H + T)$ *to* $\mathbb{C}$ *such that when* $I_H + T$ *is invertible, the canonical isomorphism from*

$\mathbb{C} = \mathrm{Det}(I_H+T)$ *to* $\mathbb{C}$ *is given by multiplying* $\det(I_H+T)^{-1}$ *to elements of* $\mathbb{C}$.

Proof. Let $R : H \to H$ be any finite-rank operator such that $\|T-R\| < 1$. Then the partial sum of the series $\sum_{k\in\mathbb{N}}(-1)^k(T-R)^k$ is a Cauchy sequence in the Banach space of bounded linear operators on H. So this series has a limit. Denote this limit by $(I_H+(T-R))^{-1}$. Then it is indeed the inverse of $I_H+(T-R)$. We have

$$I_H + T = (I_H + (T-R))(I_H + (I_H + (T-R))^{-1}R).$$

Since $I_H+(T-R)$ is invertible, by definition $\mathrm{Det}(I_H+(T-R)) = \mathbb{C}$. Since R is of finite rank, $(I_H + (T-R))^{-1}R$ is also of finite rank. By Corollary 1.2.5,

$$\mathrm{Det}(I_H + (I_H + (T-R))^{-1}R) \simeq \mathbb{C}.$$

By Corollary 1.2.4,

$$\begin{aligned}\mathrm{Det}(I_H+T) &= \mathrm{Det}((I_H + (T-R))(I_H + (I_H + (T-R))^{-1}R)) \\ &\simeq \mathrm{Det}(I_H + (T-R)) \otimes \mathrm{Det}(I_H + (I_H + (T-R))^{-1}R) \\ &\simeq \mathbb{C}. \qquad (18)\end{aligned}$$

So we obtain a map from the set $\mathcal{V}_1$ of all finite-rank operators R on H satisfying $\|T-R\| < 1$ to the space of isomorphisms from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$. We want to show that when R goes to T, the image of R under this map has a limit. We prove this existence by showing that when two finite-rank operators R_1 and R_2 are very close to T, their images under the map above are very close to each other.

Given any finite-rank operators R_1 and R_2 satisfying $\|T-R_1\| < 1$ and $\|T-R_2\| < 1$, respectively, we want to compare the isomorphisms from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$ obtained using R_1 and R_2. Since we can identify both $\mathrm{Det}(I_H + (T-R_1))$ and $\mathrm{Det}(I_H + (T-R_2))$ with $\mathbb{C}$, we need only compare the canonical isomorphisms from $\mathrm{Det}(I_H + (I_H + (T-R_1))^{-1}R_1)$ and from $\mathrm{Det}(I_H + (I_H + (T-R_2))^{-1}R_2)$ to $\mathbb{C}$. Write

$$\begin{aligned}&I_H + (I_H + (T-R_1))^{-1}R_1 \\ &\quad = (I_H + (I_H + (T-R_1))^{-1}(R_1-R_2))(I_H + (I_H + (T-R_2))^{-1}R_2).\end{aligned}$$

Then by Proposition 1.2.4, we have

$$\begin{aligned}&\mathrm{Det}(I_H + (I_H + (T-R_1))^{-1}R_1) \simeq \\ &\quad \simeq \mathrm{Det}(I_H + (I_H + (T-R_1))^{-1}(R_1-R_2)) \\ &\qquad \otimes \mathrm{Det}(I_H + (I_H + (T-R_2))^{-1}R_2).\end{aligned}$$

When $\|T-R_1\|$ and $\|T-R_2\|$ are small, $\|R_2-R_1\|$ is small. When $\|R_2-R_1\|$ is sufficiently small,

$$I_H+(I_H+(T-R_1))^{-1}(R_1-R_2)$$

is invertible and

$$\mathrm{Det}(I_H+(I_H+(T-R_1))^{-1}(R_1-R_2))=\mathbb{C}.$$

Thus we have a canonical isomorphism from

$$V_2=\mathrm{Det}(I_H+(I_H+(T-R_2))^{-1}R_2)$$

to

$$V_1=\mathrm{Det}(I_H+(I_H+(T-R_1))^{-1}R_1)$$

and the composition of this isomorphism with the canonical isomorphism from V_1 to $\mathbb{C}$ is equal to the canonical isomorphism from V_2 to $\mathbb{C}$ multiplied by

$$\det{}_{12}=\det(I_H+(I_H+(T-R_1))^{-1}(R_1-R_2)).$$

By Proposition 1.2.6, (18) and the discussion above, the isomorphisms from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$ obtained using (18) with R replaced by R_1 and R_2 differ from each other by the factor $\det_{12}$. Since $\det(I_H+T)$ exists, $\det{}_{12}$ is very small if both R_1 and R_2 are very close to T. Since the space of isomorphisms from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$ is complete, there is an isomorphism from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$ such that when R goes to T, the isomorphism from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$ obtained (18) is convergent to this isomorphism. This isomorphism is the canonical isomorphism from $\mathrm{Det}(I_H+T)$ to $\mathbb{C}$.

Now assume that I_H+T is invertible. Then there is a complete orthonormal basis $\{\psi_n\}_{n\in\mathbb{Z}_+}$ of H such that

$$T=\sum_{n\in\mathbb{Z}_+}\lambda_n(\psi_n,\cdot)\psi_n,$$

where $(\cdot,\cdot)$ is the inner product in H and the equality means that the right-hand side is convergent strongly to the left-hand side. We choose

$$R_n=\sum_{k=1}^{n}\lambda_k(\psi_k,\cdot)\psi_k,\quad n\in\mathbb{Z}_+.$$

Then $(T-R_n)|_{\mathrm{Im}R_n}=0$. Note that when n is sufficiently large, $I_H+(T-R_n)$ is invertible and we have

$$(I_H+(T-R_n))^{-1}=\sum_{m\in\mathbb{N}}(-1)^m(T-R_n)^m.$$

Thus

$$\begin{aligned}(I_H + (T - R_n))^{-1}|_{\mathrm{Im}R_n} &= \sum_{m\in\mathbb{N}} (-1)^m (T - R_n)^m|_{\mathrm{Im}R_n} \\ &= I_{\mathrm{Im}R_n}.\end{aligned}$$

So for sufficiently large n, we have

$$(I_H + (T - R_n))^{-1} R_n = I_{\mathrm{Im}R_n} R_n = R_n.$$

Since $I_H + T$ is invertible,

$$I_H + R_n = I_H + (I_H + (T - R_n))^{-1} R_n = (I_H + (T - R_n))^{-1}(I_H + T)$$

is also invertible. By Proposition 1.2.5, we know that the canonical isomorphism from

$$\mathbb{C} = \mathrm{Det}(I_H + (I_H + (T - R_n))^{-1} R_n)$$

to $\mathbb{C}$ is given by multiplying the elements of $\mathbb{C}$ by

$$\det(I_H + R_n)^{-1} = \left(\prod_{k=1}^{n} (1 + \lambda_k)\right)^{-1}.$$

On the other hand, the canonical isomorphism from $\mathbb{C} = \mathrm{Det}(I_H + T)$ to

$$\mathbb{C} = \mathrm{Det}(I_H + (I_H + (T - R_n))^{-1} R_n)$$

in this case is the trivial map. So we see that the canonical isomorphism from $\mathbb{C} = \mathrm{Det}(I_H + T)$ to $\mathbb{C}$ given by R_n is also given by multiplying the elements of $\mathbb{C}$ by

$$\det(I_H + R_n)^{-1}.$$

Thus the limit of these isomorphisms is given by multiplying the elements of $\mathbb{C}$ by

$$\left(\prod_{k\in\mathbb{Z}_+} (1 + \lambda_k)\right)^{-1} = (\det(I_H + T))^{-1} = \det(I_H + T)^{-1}.$$

□

Proposition 1.2.10. *Let $F : H_1 \to H_2$ be a Fredholm operator and G a quasi-inverse of F such that $FG = I_{H_2} + T$ where T is of trace class. Then $\mathrm{Det}F \simeq (\mathrm{Det}G)^*$ such that in the case that both F and G are invertible, the canonical isomorphism from $\mathrm{Det}F = \mathbb{C}$ to $(\mathrm{Det}G)^* = \mathbb{C}$ is given by multiplying $\det(FG)^{-1} = (\det(I_{H_2} + T))^{-1}$ to elements of $\mathbb{C}$.*

PROOF. By Proposition 1.2.9, $\mathrm{Det}(I_{H_2}+T)\simeq\mathbb{C}$. By Proposition 1.2.4, $\mathrm{Det}FG\simeq\mathrm{Det}F\otimes\mathrm{Det}G$. Thus $\mathrm{Det}F\otimes\mathrm{Det}G\simeq\mathbb{C}$, or equivalently, $\mathrm{Det}F\simeq(\mathrm{Det}G)^*$.

If both F and G are invertible, $I_{H_2}+T=FG$ is also invertible. By Proposition 1.2.9, the canonical isomorphism from

$$\mathrm{Det}FG=\mathrm{Det}(I_{H_2}+T)=\mathbb{C}$$

to $\mathbb{C}$ is given by multiplying

$$\det(FG)^{-1}=(\det(I_{H_2}+T))^{-1}$$

to the elements of $\mathbb{C}$. Thus the induced canonical isomorphism from $\mathrm{Det}F=\mathbb{C}$ to $(\mathrm{Det}G)^*=\mathbb{C}$ is also given by multiplying

$$\det(FG)^{-1}=(\det(I_{H_2}+T))^{-1}$$

to elements of $\mathbb{C}$. □

COROLLARY 1.2.11. *Let $F:H_1\to H_2$ be a Fredholm operator and $T:H_1\to H_2$ a trace-class operator. Then there is a canonical isomorphism from* $\mathrm{Det}(F+T)$ *to* $\mathrm{Det}F$ *such that in the case that both F and $F+T$ are invertible, $(F+T)F^{-1}$ has a determinant and the canonical isomorphism from* $\mathrm{Det}(F+T)=\mathbb{C}$ *to* $\mathrm{Det}F=\mathbb{C}$ *is given by multiplying* $(\det((F+T)F^{-1}))^{-1}$ *to elements of* $\mathbb{C}$. *Moreover, if $F:H_1\to H_2$ is a Fredholm operator and $T,T':H_1\to H_2$ are trace-class operators, then the diagram*

$$\begin{array}{ccc}\mathrm{Det}(F+T+T') & \longrightarrow & \mathrm{Det}(F+T)\\ \downarrow & & \downarrow\\ \mathrm{Det}(F+T') & \longrightarrow & \mathrm{Det}F\end{array}\tag{19}$$

is commutative.

PROOF. Let G be the canonical quasi-inverse of F in Remark 1.1.5. Then G is also a quasi-inverse of $F+T$ such that $(F+T)G-I_{H_2}$ and $G(F+T)-I_{H_1}$ are trace-class operators. By Proposition 1.2.10,

$$\mathrm{Det}(F+T)\simeq(\mathrm{Det}G)^*\simeq\mathrm{Det}F.$$

If both F and $F+T$ are invertible,

$$(F+T)F^{-1}=I_{H_1}+TF^{-1}$$

and the operator TF^{-1} is of trace class. So $(F+T)F^{-1}$ has a determinant. In this case, the canonical quasi-inverse is $G=F^{-1}$. Thus by Proposition 1.2.10, the canonical isomorphism from $\mathrm{Det}(F+T)=\mathbb{C}$ to $(\mathrm{Det}G)^*$ is given by multiplying $(\det((F+T)F^{-1}))^{-1}$ to elements of $\mathbb{C}$ and the canonical isomorphism from $(\mathrm{Det}G)^*=\mathbb{C}$ to $\mathrm{Det}F=\mathbb{C}$ is the identity operator on $\mathbb{C}$. The conclusion we need follows immediately.

To prove (19), we need only prove that the diagram

$$\begin{array}{ccccc} \mathrm{Det}(F+T+T')G & \longrightarrow & \mathbb{C} & \longrightarrow & \mathrm{Det}(F+T)G \\ \downarrow & & & & \downarrow \\ \mathbb{C} & & & & \mathbb{C} \\ \downarrow & & & & \downarrow \\ \mathrm{Det}(F+T')G & \longrightarrow & \mathbb{C} & \longrightarrow & \mathrm{Det}FG \end{array} \tag{20}$$

is commutative. But both paths in (20) are equal to the composition of the canonical isomorphism from $\mathrm{Det}(F+T+T')G$ to $\mathbb{C}$ and the canonical isomorphism from $\mathbb{C}$ to $\mathrm{Det}FG$. So (20) is commutative. $\square$

Proposition 1.2.12. *Let H_i, $i = 1, \ldots, 5$, be Hilbert spaces, $F_1 : H_1 \to H_4$ and $F_2 : H_2 \to H_5$ Fredholm operators and $T_1 : H_1 \to H_2$ and $T_2 : H_2 \to H_5$ trace-class operators. If the diagrams*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H_1 & \longrightarrow & H_2 & \longrightarrow & H_3 & \longrightarrow & 0 \\ & & {\scriptstyle F_1}\downarrow & & {\scriptstyle F_2}\downarrow & & {\scriptstyle I_{H_3}}\downarrow & & \\ 0 & \longrightarrow & H_4 & \longrightarrow & H_5 & \longrightarrow & H_3 & \longrightarrow & 0 \end{array} \tag{21}$$

and

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H_1 & \longrightarrow & H_2 & \longrightarrow & H_3 & \longrightarrow & 0 \\ & & {\scriptstyle F_1+T_1}\downarrow & & {\scriptstyle F_2+T_2}\downarrow & & {\scriptstyle I_{H_3}}\downarrow & & \\ 0 & \longrightarrow & H_4 & \longrightarrow & H_5 & \longrightarrow & H_3 & \longrightarrow & 0 \end{array} \tag{22}$$

have same exact rows and are commutative, then the diagram

$$\begin{array}{ccc} \mathrm{Det}(F_2+T_2) & \longrightarrow & \mathrm{Det}(F_1+T_1) \\ \downarrow & & \downarrow \\ \mathrm{Det}F_2 & \longrightarrow & \mathrm{Det}F_1 \end{array} \tag{23}$$

is commutative.

Proof. Since (21) and (22) have exact rows and are commutative, we can embed H_1 and H_4 into H_2 and H_5, respectively and after the

embedding,

$$\begin{aligned} F_1 &= F_2|_{H_1}, \\ F_1 + T_1 &= (F_2 + T_2)|_{H_1}, \\ \mathrm{Ker} F_1 &= \mathrm{Ker} F_2, \\ \mathrm{Ker}(F_1 + T_1) &= \mathrm{Ker}(F_2 + T_2), \\ \mathrm{Im} F_1 &= \mathrm{Im} F_2 \cap H_4, \\ \mathrm{Im}(F_1 + T_1) &= \mathrm{Im}(F_2 + T_2) \cap H_4. \end{aligned}$$

In particular, we can identify $\mathrm{Coker} F_1$ and $\mathrm{Coker}(F_1+T_1)$ with $\mathrm{Coker} F_2$ and $\mathrm{Coker}(F_2 + T_2)$, respectively. Thus after the embedding and the identification above,

$$\mathrm{Det} F_1 = \mathrm{Det} F_2, \tag{24}$$

$$\mathrm{Det}(F_1 + T_1) = \mathrm{Det}(F_2 + T_2). \tag{25}$$

To prove the commutativity of (23), we need only prove that the canonical isomorphisms from (25) to (24) obtained from $\mathrm{Det}(F_1+T_1) \simeq \mathrm{Det} F_1$ and from $\mathrm{Det}(F_2 + T_2) \simeq \mathrm{Det} F_2$ are the same. But from the proofs of Propositions 1.2.5, 1.2.9, 1.2.10 and Corollary 1.2.11, it is clear that these two canonical isomorphisms are equal. □

1.3. Determinant lines over Riemann surfaces with parametrized boundaries

We construct the determinant lines over Riemann surfaces with parametrized boundaries and discuss their basic properties in this section.

In this section, a *Riemann surface* means a one-dimensional compact connected complex manifold and a *Riemann surface with boundary* means a one-dimensional compact connected complex manifold with boundary. In the next section we shall also consider disjoint unions of connected Riemann surfaces.

It is clear that the boundary of a one-dimensional compact connected complex manifold with boundary is a one-dimensional compact real manifold and thus is the union of its connected components, each of which is diffeomorphic to the circle. If the boundary has infinitely many connected components, any infinite sequence whose components belong to different connected components of the boundary must has a convergent subsequence since the complex manifold is compact, contradicting the fact that the boundary is a manifold. Thus there can only be finitely many connected components.

An orientation of a boundary component of a Riemann surface with boundary is said to be *positive* (*negative*) if the orientation is the same as (opposite to) the orientation induced from the orientation of the Riemann surface with boundary determined by its complex structure. A bijective map from S^1 to a boundary component of a Riemann surface with boundary is called an *analytic parametrization* of the boundary component if there exists a real number $r \neq 1$ such that the map can be extended to an analytic map from

$$\mathbb{A}_r = \{z \in \mathbb{C} \mid r \leq |z| \leq 1\}$$

when $r < 1$ or from

$$\mathbb{A}_r = \{z \in \mathbb{C} \mid 1 \leq |z| \leq r\}$$

when $r > 1$ to the Riemann surface with boundary. An analytic parametrization of a boundary component of a Riemann surface with boundary determine an orientation of the boundary component: It is the push-forward of the orientation on S^1 induced from the orientation of $\mathbb{A}_r$. In the case $r < 1$, the orientation determined by the parametrization is positive and in the case $r > 1$, the orientation determined by the parametrization is negative.

A *Riemann surface with parametrized boundary* is a Riemann surface with boundary whose boundary components are equipped with analytic parametrizations and both the sets of positively and negatively oriented boundary components are ordered. By the discussion above, we see that the boundary components of any Riemann surface with analytically parametrized boundary components have orientations determined by the parametrizations.

Let Σ be a Riemann surface with parametrized boundary. We have the Cauchy-Riemann operator $\overline{\partial}$ from the space $\Omega^0(\Sigma) = \Omega^{0,0}(\Sigma)$ of smooth functions on the surface to the space $\Omega^{0,1}(\Sigma)$ of $(0,1)$-forms on the surface. The boundary of Σ can be decomposed as $\partial\Sigma = \cup_{i=1}^k C_i^{\epsilon_i}$ where for any i, $1 \leq i \leq k$, $C_i^{\epsilon_i}$ is a connected component of $\partial\Sigma$ and $\epsilon_i = \pm$ indicates the orientation of the component. By definition, for any i, $1 \leq i \leq k$, $C_i^{\epsilon_i}$ is parametrized analytically by a map from the circle S^1 to $C_i^{\epsilon_i}$. Any smooth function on $C_i^{\epsilon_i}$ can be decomposed as the sum of two smooth functions, one of which, as a function on S^1, has a Fourier expansion of the form $\sum_{n\in\mathbb{N}} a_n e^{2\pi n\theta i}$ (θ is the usual parametrization of the circle by angles) and the other of which, as a function on S^1, has a Fourier expansion of the form $\sum_{n\in-\mathbb{Z}_+} a_n e^{2\pi n\theta i}$. If $\epsilon_i = +$ ($\epsilon_i = -$), that is, this component is positively (negatively) oriented, we denote by $\Omega_+^0(C_i^{\epsilon_i})$ the space of all smooth functions on $C_i^{\epsilon_i}$ which, as functions on S^1, have Fourier expansions of the form

$\sum_{n\in\mathbb{N}} a_n e^{2\pi n\theta i}$ ($\sum_{n\in-\mathbb{Z}_+} a_n e^{2\pi n\theta i}$) and by $\Omega^0_-(C_i^{\epsilon_i})$ the space of smooth functions on $C_i^{\epsilon_i}$ which, as functions on S^1, have Fourier expansions of the form $\sum_{n\in-\mathbb{Z}_+} a_n e^{2\pi n\theta i}$ ($\sum_{n\in\mathbb{N}} a_n e^{2\pi n\theta i}$). Thus the space $\Omega^0(\partial\Sigma)$ of all smooth functions on $\partial\Sigma$ can be decomposed as $\oplus_{i=1}^k(\Omega^0_+(C_i^{\epsilon_i}) \oplus \Omega^0_-(C_i^{\epsilon_i}))$. Let

$$\Omega^0_\pm(\partial\Sigma) = \oplus_{i=1}^k \Omega^0_{\pm\epsilon_i}(C_i^{\epsilon_i}) \subset \Omega^0(\partial\Sigma)$$

and let pr be the composition of the restriction from $\Omega^0(\Sigma)$ to $\Omega^0(\partial\Sigma)$ and the projection from $\Omega^0(\partial\Sigma)$ to $\Omega^0_+(\partial\Sigma)$. We have an operator

$$\overline{\partial} \oplus \mathrm{pr} : \Omega^0(\Sigma) \to \Omega^{0,1}(\Sigma) \oplus \Omega^0_+(\partial\Sigma).$$

We now want to show that $\bar\partial \oplus \mathrm{pr}$ can be regarded as a Fredholm operator. Since both $\Omega^0(\Sigma)$ and $\Omega^{0,1}(\Sigma) \oplus \Omega^0_+(\partial\Sigma)$ are not Hilbert spaces, we have to extend these spaces and the operator in order to discuss whether we have a Fredholm operator. We shall use results in the theory of elliptic boundary problems. Let Σ° be the interior of Σ, $\bar H_{(s)}(\Sigma^\circ)$, $s \in \mathbb{R}$, the Sobolev spaces on Σ° defined in [**?**] and $H_{(s)}(\partial\Sigma)$, $s \in \mathbb{R}$, the Sobolev spaces on $\partial\Sigma$. For convenience, we shall denote $\bar H_{(s)}(\Sigma^\circ)$ and $\bar H_{(s)}(\Sigma^\circ) \otimes \Omega^{0,1}(\Sigma)$ simply as $H_{(s)}(\Sigma)$ and $\Omega^{0,1}_{(s)}(\Sigma)$, respectively. Let

$$R : \Omega^0(\Sigma) \to \Omega^0(\partial\Sigma)$$

be the operator taking a function on $\Omega^0(\Sigma)$ to its restriction on $\Omega^0(\partial\Sigma)$. Then we have

Proposition 1.3.1. *The operator $\bar\partial \oplus R$ can be extended uniquely to a Fredholm operator from $H_{(s)}(\Sigma)$ to*

$$\Omega^{0,1}_{(s-1)}(\Sigma) \oplus H_{(s-\frac{1}{2})}(\partial\Sigma) \tag{26}$$

for any $s \geq 1$. The kernel of this Fredholm operator is in $\Omega^0(\Sigma)$ and the range is the orthogonal space of a finite-dimensional subspace of $\Omega^{0,1}(\Sigma) \oplus \Omega^0(\partial\Sigma)$ in (26). In particular, the kernel of this extension is equal to the kernel of $\bar\partial \oplus R$ and the cokernel of this extension is canonically isomorphic to the cokernel of $\bar\partial \oplus R$.

Proof. It easy to verify that the Dirichlet boundary problem

$$\bar\partial f = g, \quad R(f) = f|_{\partial\Sigma} = h, \quad g \in \Omega^{0,1}(\Sigma), \quad h \in \Omega^0(\partial\Sigma)$$

is elliptic (see, for example, [**?**] for the definition of elliptic boundary problem). Thus by Theorems 20.1.2 and 20.1.8 in [**?**], $\bar\partial \oplus R$ can be extended to a Fredholm operator from $H_{(s)}(\Sigma)$ to (26) for any $s \geq 1$ having the properties stated in the proposition. □

For any $s \in \mathbb{R}_+$, let $H^+_{(s-\frac{1}{2})}(\partial\Sigma)$ and $H^-_{(s-\frac{1}{2})}(\partial\Sigma)$ be the completion of $\Omega^0_+(\partial\Sigma)$ and $\Omega^0_-(\partial\Sigma)$, respectively, in $H_{(s-\frac{1}{2})}(\partial\Sigma)$.

COROLLARY 1.3.2. *The operator* $\bar{\partial} \oplus \mathrm{pr}$ *can be extended uniquely to a Fredholm operator from* $H_{(s)}(\Sigma)$ *to*

$$\Omega^{0,1}_{(s-1)}(\Sigma) \oplus H^+_{(s-\frac{1}{2})}(\partial\Sigma) \tag{27}$$

for any $s \geq 1$. *The kernel of this Fredholm operator is* 0 *if* Σ *has at least one positively oriented boundary component and is the space of constant functions if* Σ *has no positively oriented boundary components. The range is the orthogonal space of a finite-dimensional subspace of* $\Omega^{0,1}(\Sigma) \oplus \Omega^0_+(\partial\Sigma)$ *in (27). In particular, the kernel of this extension is equal to the kernel of* $\bar{\partial} \oplus \mathrm{pr}$ *and the cokernel of this extension is canonically isomorphic to the cokernel of* $\bar{\partial} \oplus \mathrm{pr}$.

PROOF. The extension of $\bar{\partial} \oplus \mathrm{pr}$ is defined to be the composition of the projection from $H_{(s-\frac{1}{2})}(\partial\Sigma)$ to $H^+_{(s-\frac{1}{2})}(\partial\Sigma)$ and the extension of $\bar{\partial} \oplus R$. Let f be an element of the kernel of this extension. Then f can be extended to an analytic function on a Riemann surface without boundary. Thus f must be a constant. If Σ has at least one positively oriented boundary component, this constant must be 0. If Σ has no positively oriented boundary components, any constant function is an element of the kernel. So the kernel of this extension is 0 if Σ has at least one positively oriented boundary component and is the space of constant functions if Σ has no positively oriented boundary components. By the definition, the dimension of the cokernel of this extension is less than or equal to the dimension of the cokernel of the extension of $\bar{\partial} \oplus R$. So this extension is a Fredholm operator. Since the range of the extension of $\bar{\partial} \oplus R$ is the orthogonal space of a finite-dimensional subspace of $\Omega^{0,1}(\Sigma) \oplus \Omega^0(\partial\Sigma)$ in (26), the range of this extension is the orthogonal space of a finite-dimensional subspace of $\Omega^{0,1}(\Sigma) \oplus \Omega^0_+(\partial\Sigma)$ in (27). Thus the kernel of this extension is equal to the kernel of $\bar{\partial} \oplus \mathrm{pr}$ and the cokernel of this extension is isomorphic to the cokernel of $\bar{\partial} \oplus \mathrm{pr}$. $\square$

We shall use the same notation $\bar{\partial} \oplus \mathrm{pr}$ to denote its extension to $H_{(s)}(\Sigma)$ for any $s \geq 1$. The determinant line

$$\mathrm{Det}(\bar{\partial} \oplus \mathrm{pr}) = \mathrm{Det}(\mathrm{Ker}(\bar{\partial} \oplus \mathrm{pr}))^* \otimes \mathrm{Det}(\mathrm{Coker}(\bar{\partial} \oplus \mathrm{pr}))$$

of $\bar{\partial} \oplus \mathrm{pr}$ is well defined.

Let $\mathrm{Hol}(\Sigma)$ be the space of all analytic functions on Σ and π_Σ the restriction of pr to $\mathrm{Hol}(\Sigma)$. We have:

PROPOSITION 1.3.3. *The operator*

$$\pi_\Sigma : \mathrm{Hol}(\Sigma) \to \Omega^0_+(\partial\Sigma)$$

can be extended to a Fredholm operator from the completion $\overline{\mathrm{Hol}}_{(s)}(\Sigma)$ *of* $\mathrm{Hol}(\Sigma)$ *in* $H_{(s)}(\Sigma)$ *to* $H^+_{(s-\frac{1}{2})}(\partial\Sigma)$ *for any* $s \geq 1$. *The kernel of this extension is in* $\mathrm{Hol}(\Sigma)$ *and the range of this extension is the orthogonal complement of a finite-dimensional subspace of* $\Omega^0_+(\partial\Sigma)$ *in* $H^+_{(s-\frac{1}{2})}(\partial\Sigma)$. *In particular, the kernel of of this extension is equal to the kernel of* π_Σ *and the cokernel of this extension is canonically isomorphic to the cokernel of* π_Σ. *Moreover, there is a canonical isomorphism from the determinant line* $\mathrm{Det}(\bar\partial \oplus \mathrm{pr})$ *to* $\mathrm{Det}\pi_\Sigma$.

PROOF. The commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathrm{Hol}(\Sigma) & \longrightarrow & \Omega^0(\Sigma) & \xrightarrow{\bar\partial} & \Omega^{0,1}(\Sigma) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \pi_\Sigma} & & \downarrow{\scriptstyle \bar\partial\oplus\mathrm{pr}} & & \downarrow{\scriptstyle I_{\Omega^{0,1}(\Sigma)}} & & \\
0 & \longrightarrow & \Omega^0_+(\partial\Sigma) & \longrightarrow & \Omega^{0,1}(\Sigma)\oplus\Omega^0_+(\partial\Sigma) & \longrightarrow & \Omega^{0,1}(\Sigma) & \longrightarrow & 0
\end{array}$$

(28)

induces the commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \overline{\mathrm{Hol}}_{(s)}(\Sigma) & \longrightarrow & H_{(s)}(\Sigma) & \xrightarrow{\bar\partial} & \Omega^{0,1}_{(s-1)}(\Sigma) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \pi_\Sigma} & & \downarrow{\scriptstyle \bar\partial\oplus\mathrm{pr}} & & \downarrow{\scriptstyle I_{\Omega^{0,1}_{(s-1)}(\Sigma)}} & & \\
0 & \longrightarrow & H^+_{(s-\frac{1}{2})}(\partial\Sigma) & \longrightarrow & \Omega^{0,1}_{(s-1)}(\Sigma)\oplus H^+_{(s-\frac{1}{2})}(\partial\Sigma) & \longrightarrow & \Omega^{0,1}_{(s-1)}(\Sigma) & \longrightarrow & 0
\end{array}$$

(29)

where we use the same notations to denote operators and their extensions. Since the rows of the diagram (28) are exact, the rows of (29) is also exact. By the diagram (29), the kernel of π_Σ is in the kernel of $\bar\partial \oplus \mathrm{pr}$ and thus is in $\mathrm{Hol}(\Sigma)$ and is finite-dimensional. Let V be the image of the finite-dimensional subspace orthogonal to the range of $\bar\partial \oplus \mathrm{pr}$ under the projection from (27) to $H^+_{(s-\frac{1}{2})}(\partial\Sigma)$. Then V is also finite-dimensional. By the diagram (29), we see that the range of π_Σ is equal to the range of the composition of the projection above and $\bar\partial\oplus\mathrm{pr}$. Thus the range of π_Σ is the orthogonal space of V in $H^+_{(s-\frac{1}{2})}(\partial\Sigma)$. So the cokernel of π_Σ is finite-dimensional and π_Σ is a Fredholm operator. Since the finite-dimensional subspace orthogonal to the range of $\bar\partial \oplus \mathrm{pr}$ is in $\Omega^{0,1}(\Sigma)) \oplus \Omega^0_+(\partial\Sigma)$, V is in $\Omega^0_+(\partial\Sigma)$. Thus the cokernel of π_Σ is canonically isomorphic to the cokernel of its extension. Finally

by (29) and Proposition 1.2.3, there is a canonical isomorphism from the determinant line $\mathrm{Det}(\bar{\partial} \oplus \mathrm{pr})$ to $\mathrm{Det}\pi_\Sigma$. $\square$

We shall call $\mathrm{Det}\pi_\Sigma$ the *determinant line over* Σ and denote it Det_Σ. It is obvious that $\Sigma \mapsto \mathrm{Det}_\Sigma$ gives a functor from the category of Riemann surfaces with parametrized boundaries to the category of one-dimensional vector spaces with isomorphisms as morphisms.

1.4. Canonical isomorphisms associated to sewing and determinant line bundles over moduli spaces

Let Σ_1 and Σ_2 be two Riemann surfaces with parametrized boundaries. Let S_1 be a subset of the set of the boundary components of Σ_1 and S_2 a subset of the set of the boundary components of Σ_2. If the number of negatively (positively) oriented boundary components in S_1 is equal to the number of positively (negatively) oriented boundary components in S_2, we say that S_1 and S_2 are *compatible for sewing*. In this case, we define $\Sigma_1 \# \Sigma_2$ to be the Riemann surface with parametrized boundary obtained by identifying the negatively and positively oriented boundary components in S_1 with the positively and negatively oriented boundary components of Σ_2, respectively, using the given orderings of the boundary components to determine how to match the components to be sewn and using the given parametrizations of the boundary components. Note that the operation $\#$ depends on S_1 and S_2. For simplicity, we shall not include this dependence explicitly in the notation $\#$. The spaces $\Omega^0_\pm(S_1)$, $\Omega^0_\pm(S_2)$, $H^\pm_{(s-\frac{1}{2})}(S_1)$ and $H^\pm_{(s-\frac{1}{2})}(S_2)$ for $s \in \mathbb{R}_+$ are defined in the obvious way.

One of the most important properties of the determinant lines defined in the preceding section is that for any Σ_1 and Σ_2 as above, there is a canonical isomorphism

$$\ell_{\Sigma_1,\Sigma_2} : \mathrm{Det}_{\Sigma_1} \otimes \mathrm{Det}_{\Sigma_2} \to \mathrm{Det}_{\Sigma_1 \# \Sigma_2}$$

satisfying a natural associativity property. The first half of this section is devoted to the construction of ℓ_{Σ_1,Σ_2} and to the proof of the associativity property.

Let S be the curve on $\Sigma_1 \# \Sigma_2$ corresponding to S_1 or S_2 and $\Sigma_1 \sqcup \Sigma_2$ the disjoint union of Σ_1 and Σ_2. Then $\Sigma_1 \sqcup \Sigma_2$ is a compact Riemann surface with parametrized boundary and with two ordered components. We define

$$\Delta_{\Sigma_1,\Sigma_2} : \mathrm{Hol}(\Sigma_1 \sqcup \Sigma_2) \to \Omega^0(S)$$

by

$$\Delta_{\Sigma_1,\Sigma_2}(f) = f|_{S_1} - f|_{S_2}$$

and

$$\bar{\pi}_{\Sigma_1,\Sigma_2} : \mathrm{Hol}(\Sigma_1 \sqcup \Sigma_2) \to \Omega^0_+(\partial(\Sigma_1\#\Sigma_2)) \oplus \Omega^0(S)$$

by

$$\bar{\pi}_{\Sigma_1,\Sigma_2}(f) = ((f|_{\partial(\Sigma_1\#\Sigma_2)})_+, \Delta_{\Sigma_1,\Sigma_2}(f))$$

where $(f|_{\partial(\Sigma_1\#\Sigma_2)})_+$ denotes the projection of

$$f|_{\partial(\Sigma_1\#\Sigma_2)} \in \Omega^0(\partial(\Sigma_1\#\Sigma_2))$$

to $\Omega^0_+(\partial(\Sigma_1\#\Sigma_2))$. Then we have a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathrm{Hol}(\Sigma_1\#\Sigma_2) & \longrightarrow & \mathrm{Hol}(\Sigma_1 \sqcup \Sigma_2) & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}} & \Omega^0(S) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \pi_{\Sigma_1\#\Sigma_2}} & & \downarrow{\scriptstyle \bar{\pi}_{\Sigma_1,\Sigma_2}} & & \downarrow{\scriptstyle I_{\Omega^0(S)}} & & \\
0 & \longrightarrow & \Omega^0_+(\partial(\Sigma_1\#\Sigma_2)) & \longrightarrow & \Omega^0_+(\partial(\Sigma_1\#\Sigma_2)) \oplus \Omega^0(S) & \longrightarrow & \Omega^0(S) & \longrightarrow & 0
\end{array}$$

(30)

Lemma 1.4.1. *The rows of the diagram (30) are exact.*

Proof. The second row of (30) is exact by definition. We prove that the first row is also exact. The embedding of $\mathrm{Hol}(\Sigma_1\#\Sigma_2)$ in $\mathrm{Hol}(\Sigma_1 \sqcup \Sigma_2)$ is by definition injective. The subspace $\mathrm{Hol}(\Sigma_1\#\Sigma_2)$ of $\mathrm{Hol}(\Sigma_1\sqcup\Sigma_2)$ is by definition in the kernel of $\Delta_{\Sigma_1,\Sigma_2}$. Let f be a function in the kernel of $\Delta_{\Sigma_1,\Sigma_2}$. Then f is continuous on $\Sigma_1\#\Sigma_2$ and is analytic on $(\Sigma_1\#\Sigma_2) \setminus S$. Using the Cauchy theorem, we see that f is also analytic on S. Thus f is in $\mathrm{Hol}(\Sigma_1\#\Sigma_2)$. This proves that the first row is exact at $\mathrm{Hol}(\Sigma_1 \sqcup \Sigma_2)$.

To show that the first row is exact, we need to show that $\Delta_{\Sigma_1,\Sigma_2}$ is surjective. Let $\overline{\Sigma_1\#\Sigma_2}$ be the compact Riemann surface without boundary obtained from $\Sigma_1\#\Sigma_2$ by sewing disks along its boundary components using their analytic parametrizations. Since when the genus of $\overline{\Sigma_1\#\Sigma_2}$, or equivalently, the genus of $\Sigma_1\#\Sigma_2$ is zero, the proof is simpler. In this work, we need only the genus-zero case and we only prove the surjectivity of $\Delta_{\Sigma_1,\Sigma_2}$ in this case.

In this genus-zero case, by the uniformization theorem (Theorem **??**), we can always find a conformal equivalence from $\overline{\Sigma_1\#\Sigma_2}$ to $\widehat{\mathbb{C}}$ such that S is mapped to a smooth simple Jordan curve and such that the interiors of Σ_1 and Σ_2 are mapped to the exterior and to the interior of the curve, respectively. Thus we need only prove the surjectivity of $\Delta_{\Sigma_1,\Sigma_2}$ when S is a smooth simple Jordan curve on the complex plane and Σ_1 and Σ_2 are the unions of S with its exterior and with its interior, respectively. In this case, for any $f \in \Omega^0(S)$, we define functions

$$g_1(z) = \int_S \frac{f(\xi)}{\xi - z} d\xi$$

when z is in the exterior of S and

$$g_2(z) = \int_S \frac{f(\xi)}{\xi - z} d\xi$$

when z is in the interior of S. Obviously g_1 and g_2 are analytic in the exterior and interior of S, respectively. Let $z_0 \in S$ and z in the interior of S. Note that if we define the value of $\frac{f(\xi)-f(z_0)}{\xi-z_0}$ at $\xi = z_0$ to be $f'(z_0)$, $\frac{f(\xi)-f(z_0)}{\xi-z_0}$ becomes a smooth function on S. We need the following result:

LEMMA 1.4.2. *When z approaches z_0 from the exterior of S along the normal line of S, $g_1(z)$ converges to*

$$\int_S \frac{f(\xi) - f(z_0)}{\xi - z_0} d\xi$$

uniformly with respect to z_0. When z approaches z_0 from the interior of S along the normal line of S, $g_2(z)$ converges to

$$f(z_0) + \int_S \frac{f(\xi) - f(z_0)}{\xi - z_0} d\xi$$

uniformly with respect to z_0. ∎

The proof of this lemma is elementary and can be found in Chapter 4, Section 3 of [**?**]. Since S is smooth, f is smooth and the convergence given in Lemma 1.4.2 is uniform with respect to $z_0 \in S$, we conclude that $\lim_{z\to z_0} g_1(z)$ and $\lim_{z\to z_0} g_2(z)$ exist and are equal to

$$\int_S \frac{f(\xi) - f(z_0)}{\xi - z_0} d\xi$$

and

$$f(z_0) + \int_S \frac{f(\xi) - f(z_0)}{\xi - z_0} d\xi,$$

respectively.

Using these limits, we define the boundary values of g_1 and g_2. The functions g_1 and g_2 are analytic in the exterior and interior of S, respectively, and are continuous on the unions of S with its exterior and with its interior, respectively, and satisfy the relation

$$g_2(z_0) - g_1(z_0) = f(z_0)$$

for $z_0 \in S$. Since

$$\frac{d}{dz_0} \int_S \frac{f(\xi) - f(z_0)}{\xi - z_0} d\xi = \int_S \left(\frac{f(\xi) - f(z_0)}{(\xi - z_0)^2} - \frac{f'(z_0)}{\xi - z_0} \right) d\xi$$

and

$$\frac{f(\xi)-f(z_0)}{(\xi-z_0)^2}-\frac{f'(z_0)}{\xi-z_0}$$

is smooth if we define its value at $\xi = z_0$ to be $f''(z_0)$, we see that g_1 and g_2 are differentiable on S. Similarly, we can show that the higher derivatives of g_1 and g_2 on S exist and thus $g_1|_S$ and $g_2|_S$ are in $\Omega^0(S)$, proving the surjectivity of $\Delta_{\Sigma_1,\Sigma_2}$ in this case. $\square$

We extend the diagram (30) to the following commutative diagram:

$$\begin{array}{ccccccc}
0 \longrightarrow & \overline{\mathrm{Hol}}_{(s)}(\Sigma_1\#\Sigma_2) & \longrightarrow & \overline{\mathrm{Hol}}_{(s)}(\Sigma_1\sqcup\Sigma_2) & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}} & & \\
 & \downarrow{\scriptstyle \pi_{\Sigma_1\#\Sigma_2}} & & \downarrow{\scriptstyle \bar{\pi}_{\Sigma_1,\Sigma_2}} & & & \\
0 \longrightarrow & H^+_{(s-\frac{1}{2})}(\partial(\Sigma_1\#\Sigma_2)) & \longrightarrow & H^+_{(s-\frac{1}{2})}(\partial(\Sigma_1\#\Sigma_2))\oplus H_{(s-\frac{1}{2})}(S) & \longrightarrow & & \\
 & & & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}} & H_{(s-\frac{1}{2})}(S) & \longrightarrow & 0 \\
 & & & & \downarrow{\scriptstyle I_{H_{(s-\frac{1}{2})}(S)}} & & \\
 & & & \longrightarrow & H_{(s-\frac{1}{2})}(S) & \longrightarrow & 0.
\end{array} \tag{31}$$

Since (30) has exact rows, (31) also has exact rows. By Proposition 1.2.3, there exists a canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2}$ to $\mathrm{Det}_{\Sigma_1\#\Sigma_2}$. (In fact, by Remark 1.2.8, to conclude $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2}\simeq\mathrm{Det}_{\Sigma_1\#\Sigma_2}$, (30) is enough.)

On the other hand, we note that the construction in the preceding section also works for Riemann surfaces that are not connected, and from the construction we see that $\mathrm{Det}_{\Sigma_1}\otimes\mathrm{Det}_{\Sigma_2}=\mathrm{Det}\pi_{\Sigma_1}\otimes\mathrm{Det}\pi_{\Sigma_2}$ is canonically isomorphic to $\mathrm{Det}_{\Sigma_1\sqcup\Sigma_2}=\mathrm{Det}\pi_{\Sigma_1\sqcup\Sigma_2}$ where $\Sigma_1\sqcup\Sigma_2$ is the disjoint union of Σ_1 and Σ_2 and $\pi_{\Sigma_1\sqcup\Sigma_2}$ is the corresponding Fredholm operator from $\overline{\mathrm{Hol}}_{(s)}(\Sigma_1\sqcup\Sigma_2)$ to $H^+_{(s-\frac{1}{2})}(\partial(\Sigma_1\sqcup\Sigma_2))$. Since

$$H^+_{(s-\frac{1}{2})}(\partial(\Sigma_1\sqcup\Sigma_2))=H^+_{(s-\frac{1}{2})}(\partial(\Sigma_1\#\Sigma_2))\oplus H^-_{(s-\frac{1}{2})}(S_1)\oplus H^+_{(s-\frac{1}{2})}(S_2)$$

and

$$H^-_{(s-\frac{1}{2})}(S_1)\oplus H^+_{(s-\frac{1}{2})}(S_2)$$

can be identified with $H_{(s-\frac{1}{2})}(S)$, we see that $\pi_{\Sigma_1\sqcup\Sigma_2}$ can also be viewed as a Fredholm operator from $\overline{\mathrm{Hol}}_{(s)}(\Sigma_1\sqcup\Sigma_2)$ to

$$H^+_{(s-\frac{1}{2})}(\partial(\Sigma_1\#\Sigma_2))\oplus H_{(s-\frac{1}{2})}(S)$$

and the difference $\pi_{\Sigma_1\sqcup\Sigma_2}-\bar{\pi}_{\Sigma_1,\Sigma_2}$ is given by

$$(\pi_{\Sigma_1\sqcup\Sigma_2}-\bar{\pi}_{\Sigma_1,\Sigma_2})(f)=(f|_{S_1})_- - (f|_{S_2})_+$$

for $f \in \overline{\mathrm{Hol}}_{(s)}(\Sigma_1 \sqcup \Sigma_2)$.

LEMMA 1.4.3. *The operator* $\pi_{\Sigma_1 \sqcup \Sigma_2} - \bar{\pi}_{\Sigma_1, \Sigma_2}$ *is a trace-class operator.*

PROOF. For simplicity, we only prove the case that S_1 contains only one negatively oriented boundary components and contains no positively oriented boundary components. Thus S_2 contains only one positively oriented boundary component and contains no negatively oriented boundary components.

We need only show that the operators given by $f \mapsto (f|_{S_1})_-$ and $f \mapsto (f|_{S_2})_+$ are of trace class. We show below that the first operator is of trace class. The proof that the second operator is of trace class is similar. Since S_1 is analytically parametrized, the parametrization of S as a map from the circle $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ to S_1 can be extended to an analytic injective map from a closed annulus $\mathbb{A}_r$ to Σ_1 for some $r > 1$. Let $\mathbb{A}$ be the image of $\mathbb{A}_r$ under this analytic map. Consider the restriction map

$$\begin{array}{rcl} P_{\mathbb{A}} : \overline{\mathrm{Hol}}_{(s)}(\Sigma_1 \sqcup \Sigma_2) & \to & \overline{\mathrm{Hol}}_{(s)}(\mathbb{A}) \\ f & \mapsto & f|_{\mathbb{A}}. \end{array}$$

Let $||\cdot||^s_{\Sigma_1 \sqcup \Sigma_2}$ and $||\cdot||^s_{\mathbb{A}}$ be the norms in $\overline{\mathrm{Hol}}_{(s)}(\Sigma_1 \sqcup \Sigma_2)$ and $\overline{\mathrm{Hol}}_{(s)}(\mathbb{A})$, respectively. Then by definition

$$||f|_{\mathbb{A}}||^s_{\mathbb{A}} \leq ||f||^s_{\Sigma_1 \sqcup \Sigma_2}$$

for any $f \in \overline{\mathrm{Hol}}_{(s)}(\Sigma_1 \sqcup \Sigma_2)$. Thus the restriction map $P_{\mathbb{A}}$ is continuous and we need only show that the map $f \mapsto (f|_{S_1})_-$ from $\overline{\mathrm{Hol}}_{(s)}(\mathbb{A})$ to $H_{(s-\frac{1}{2})}(S_1)$ is of trace class. Since $\mathbb{A}$ is analytically isomorphic to the annulus $\mathbb{A}_r$ and S_1 corresponds to the unit circle $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, we need only show that the map

$$\begin{array}{rcl} P_- : \overline{\mathrm{Hol}}_{(s)}(\mathbb{A}_r) & \to & H_{(s-\frac{1}{2})}(S^1) \\ f & \to & (f|_{S^1})_- \end{array}$$

is of trace class.

For simplicity, we only prove this fact in the case $s = 1$. The other cases can be discussed similarly. In fact, since determinant lines are independent of s, this case is enough for our purpose. Let f be an analytic function on $\mathbb{A}_r$. Then $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$ when $1 \leq |z| \leq r$. Thus for any $f \in \overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)$, we have $f = \sum_{n \in \mathbb{Z}} a_n z^n$ in $\overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)$.

We choose the inner product in $\overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)$ to be the one defined by

$$(f,g)_{\overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)} = \int_0^{2\pi}\int_1^r \bar{f}g\rho d\rho d\theta + \int_0^{2\pi}\int_1^r \frac{d\bar{f}}{d\rho}\frac{dg}{d\rho}\rho d\rho d\theta + \int_0^{2\pi}\int_1^r \frac{d\bar{f}}{d\theta}\frac{dg}{d\theta}\rho d\rho d\theta,$$

where we already write z as $\rho e^{i\theta}$, for $f,g \in \overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)$. With respect to this inner product, z^n, $n \in \mathbb{Z}$, are orthogonal to each other and

$$(z^n, z^n)_{\overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)} = \frac{\pi}{n+1}(n^2+1)(r^{2n+2}-1) + \pi n(r^{2n}-1), \quad n \in \mathbb{Z}.$$

Denote

$$\eta_n = \sqrt{\frac{\pi}{n+1}(n^2+1)(r^{2n+2}-1) + \pi n(r^{2n}-1)}, \quad n \in \mathbb{Z}.$$

Then $\left\{\frac{z^n}{\eta_n}\right\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $\overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)$.

On the other hand, the space $H_{(\frac{1}{2})}(S^1)$ has a basis $\{e^{in\theta}\}_{n\in\mathbb{Z}}$ since smooth functions on S^1 can be expanded as Fourier series. Let $f,g \in H_{(\frac{1}{2})}(S^1)$. Then we have $f = \sum_{n\in\mathbb{Z}} a_n e^{in\theta}$ and $g = \sum_{n\in\mathbb{Z}} b_n e^{in\theta}$. We choose the inner product in $H_{(\frac{1}{2})}(S^1)$ to be the one defined by

$$(f,g)_{H_{(\frac{1}{2})}(S^1)} = \sum_{n\in\mathbb{Z}}(1+|n|)\bar{a}_n b_n.$$

(It is an easy exercise to show that this definition of the inner product is equivalent to the usual definition using a cover and a partition of unity of S^1.) With respect to this inner product, $e^{in\theta}$, $n \in \mathbb{Z}$, are orthogonal to each other and

$$(e^{in\theta}, e^{in\theta})_{H_{(\frac{1}{2})}(S^1)} = 1+|n|, \quad n \in \mathbb{Z}.$$

Thus $\left\{\frac{e^{in\theta}}{\sqrt{1+|n|}}\right\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $H_{(\frac{1}{2})}(S^1)$.

For any $f = \sum_{n\in\mathbb{Z}} a_n z^n \in \overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)$, by definition

$$P_-(f) = \sum_{n\in\mathbb{Z}_+} a_n e^{in\theta} \in H_{(\frac{1}{2})}(S^1).$$

Thus

$$P_-(f) = \sum_{n\in\mathbb{N}} \frac{\sqrt{1+n}}{\eta_n}\left(\frac{z^n}{\eta_n}, f\right)_{\overline{\mathrm{Hol}}_{(1)}(\mathbb{A}_r)} \frac{e^{in\theta}}{\sqrt{1+n}}. \tag{32}$$

Since $r > 1$, $\sum_{n\in\mathbb{N}} \frac{\sqrt{1+n}}{\eta_n} < \infty$. So by (32), P_- is a trace-class operator. $\square$

Since $\pi_{\Sigma_1\sqcup\Sigma_2} - \bar{\pi}_{\Sigma_1,\Sigma_2}$ is of trace class, $\pi_{\Sigma_1\sqcup\Sigma_2}$ is the sum of $\bar{\pi}_{\Sigma_1,\Sigma_2}$ and a trace-class operator. By Corollary 1.2.11, there exists a canonical isomorphism from $\mathrm{Det}\pi_{\Sigma_1\sqcup\Sigma_2}$ to $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2}$.

Composing the canonical isomorphisms from

$$\mathrm{Det}_{\Sigma_1} \otimes \mathrm{Det}_{\Sigma_2} = \mathrm{Det}\pi_{\Sigma_1\sqcup\Sigma_2}$$

to $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2}$ and from $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2}$ to $\mathrm{Det}\pi_{\Sigma_1\#\Sigma_2}$ above, we obtain a canonical isomorphism

$$\ell_{\Sigma_1,\Sigma_2} : \mathrm{Det}_{\Sigma_1} \otimes \mathrm{Det}_{\Sigma_2} \to \mathrm{Det}_{\Sigma_1\#\Sigma_2}$$

and we call such a canonical isomorphism a *canonical isomorphism associated to sewing*.

Let Σ_1, Σ_2 and Σ_3 be Riemann surfaces with parametrized boundaries and let S_1, $S_2^{(1)}$, $S_2^{(3)}$ and S_3 be subsets of the sets of boundary components of Σ_1, Σ_2, Σ_2 and Σ_3, respectively, such that S_1 and $S_2^{(1)}$, $S_2^{(3)}$ and S_3 are compatible for sewing, and $S_2^{(1)}$ and $S_2^{(3)}$ are disjoint. Then we have Riemann surfaces with parametrized boundaries $\Sigma_1\#\Sigma_2$ $\Sigma_2\#\Sigma_3$, $\Sigma_1\#(\Sigma_2\#\Sigma_3)$, $(\Sigma_1\#\Sigma_2)\#\Sigma_3$. Furthermore, we have

$$\Sigma_1\#(\Sigma_2\#\Sigma_3) = (\Sigma_1\#\Sigma_2)\#\Sigma_3$$

where $=$ means conformally equivalent. Therefore we can choose one of them and denote it simply by $\Sigma_1\#\Sigma_2\#\Sigma_3$.

THEOREM 1.4.4. *The diagram*

$$\begin{array}{ccc} \mathrm{Det}_{\Sigma_1} \otimes \mathrm{Det}_{\Sigma_2} \otimes \mathrm{Det}_{\Sigma_3} & \xrightarrow{\ell_{\Sigma_1,\Sigma_2}\otimes I\mathrm{Det}_{\Sigma_3}} & \mathrm{Det}_{\Sigma_1\#\Sigma_2} \otimes \mathrm{Det}_{\Sigma_3} \\ {}^{I}\mathrm{Det}_{\Sigma_1}{}^{\otimes\ell_{\Sigma_2,\Sigma_3}}\Big\downarrow & & \ell_{\Sigma_1\#\Sigma_2,\Sigma_3}\Big\downarrow \\ \mathrm{Det}_{\Sigma_1} \otimes \mathrm{Det}_{\Sigma_2\#\Sigma_3} & \xrightarrow{\ell_{\Sigma_1,\Sigma_2\#\Sigma_3}} & \mathrm{Det}_{\Sigma_1\#\Sigma_2\#\Sigma_3} \end{array} \tag{33}$$

is commutative.

PROOF. Since ℓ_{Σ_1,Σ_2} for any Riemann surfaces with parametrized boundaries Σ_1 and Σ_2 is the composition of two isomorphisms, to prove the commutativity of (33), we need prove the commutativity of four diagrams. On $\Sigma_1\#\Sigma_2$ and $\Sigma_1\#\Sigma_2\#\Sigma_3$, S_1 and $S_2^{(1)}$ are identified to become a one-dimensional submanifold S_{12} of $\Sigma_1\#\Sigma_2$ and $\Sigma_1\#\Sigma_2\#\Sigma_3$, and on $\Sigma_2\#\Sigma_3$ and $\Sigma_1\#\Sigma_2\#\Sigma_3$, $S_2^{(3)}$ and S_3 are identified to become a one-dimensional submanifold S_{23} of $\Sigma_2\#\Sigma_3$ and $\Sigma_1\#\Sigma_2\#\Sigma_3$. Let

$$\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} : \mathrm{Hol}(\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3) \to \Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)) \oplus \Omega^0(S_{12}) \oplus \Omega^0(S_{23})$$

be defined by

$$\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}(f) = ((f|_{\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)})_+, \Delta_{\Sigma_1,\Sigma_2}(f), \Delta_{\Sigma_2,\Sigma_3}(f))$$

where $(f|_{\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)})_+$ is the projection of

$$f|_{\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)} \in \Omega^0(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3))$$

to $\Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3))$. Then we have

$$\begin{aligned}\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} &= \bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3} + (\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} - \bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3}) \\ &= \bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} + (\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} - \bar{\pi}_{\Sigma_1,\Sigma_2\sqcup\Sigma_3}) \\ &\quad + (\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} - \bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3}).\end{aligned} \tag{34}$$

Let

$$\kappa^{(1)}_{\Sigma_1,\Sigma_2,\Sigma_3} = \pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} - \bar{\pi}_{\Sigma_1,\Sigma_2\sqcup\Sigma_3}$$

and

$$\kappa^{(2)}_{\Sigma_1,\Sigma_2,\Sigma_3} = \pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} - \bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3}.$$

Then by (34) and Proposition 1.2.11, the diagram

$$\begin{array}{ccc} \mathrm{Det}\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3} & \longrightarrow & \mathrm{Det}(\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} + \kappa^{(1)}_{\Sigma_1,\Sigma_2,\Sigma_3}) \\ \downarrow & & \downarrow \\ \mathrm{Det}(\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} + \kappa^{(2)}_{\Sigma_1,\Sigma_2,\Sigma_3}) & \longrightarrow & \mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} \end{array} \tag{35}$$

is commutative. From definition and the surjectivity of $\Delta_{\Sigma_1,\Sigma_2}$, the diagrams

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hol}(\Sigma_1\#\Sigma_2)\oplus\mathrm{Hol}(\Sigma_3) & \longrightarrow & \mathrm{Hol}(\Sigma_1\sqcup\Sigma_2)\oplus\mathrm{Hol}(\Sigma_3) & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}\oplus 0} & \\ & & \downarrow{\scriptstyle \pi_{\Sigma_1\#\Sigma_2}\oplus\pi_{\Sigma_3}} & & \downarrow{\scriptstyle \bar{\pi}_{\Sigma_1,\Sigma_2}\oplus\pi_{\Sigma_3}} & & \\ 0 & \longrightarrow & \Omega^0_+(\partial(\Sigma_1\#\Sigma_2))\oplus\Omega^0_+(\partial\Sigma_3) & \longrightarrow & \Omega^0_+(\partial(\Sigma_1\#\Sigma_2))\oplus\Omega^0_+(\partial\Sigma_3)\oplus\Omega^0(S_{12}) & \longrightarrow & \end{array} \tag{36}$$

$$\begin{array}{ccccc} \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}\oplus 0} & \Omega^0(S_{12}) & \longrightarrow & 0 \\ & \downarrow{\scriptstyle I_{\Omega^0(S_{12})}} & & \\ \longrightarrow & \Omega^0(S_{12}) & \longrightarrow & 0 \end{array}$$

and

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hol}(\Sigma_1\#(\Sigma_2\sqcup\Sigma_3)) & \longrightarrow & \mathrm{Hol}(\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3) & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}} & \\ & & \downarrow{\scriptstyle \bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}} & & \downarrow{\scriptstyle \bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}} & & \\ 0 & \longrightarrow & \Omega^0_+(\partial(\Sigma_1\#(\Sigma_2\sqcup\Sigma_3))) & \longrightarrow & \Omega^0_+(\partial(\Sigma_1\#(\Sigma_2\sqcup\Sigma_3)))\oplus\Omega^0(S_{12}) & \longrightarrow & \end{array} \tag{37}$$

$$\begin{array}{ccccc} \xrightarrow{\Delta_{\Sigma_1,\Sigma_2}} & \Omega^0(S_{12}) & \longrightarrow & 0 \\ & \downarrow{\scriptstyle I_{\Omega^0(S_{12})}} & & \\ \longrightarrow & \Omega^0(S_{12}) & \longrightarrow & 0 \end{array}$$

have exact rows and are commutative. But $\mathrm{Hol}(\Sigma_1\#(\Sigma_2\sqcup\Sigma_3))$, $\mathrm{Hol}(\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3)$ and $\Omega^0_+(\partial(\Sigma_1\#(\Sigma_2\sqcup\Sigma_3)))$ can be identified with $\mathrm{Hol}(\Sigma_1\#\Sigma_2)\oplus\mathrm{Hol}(\Sigma_3)$, $\mathrm{Hol}(\Sigma_1\sqcup\Sigma_2)\oplus\mathrm{Hol}(\Sigma_3)$ and $\Omega^0_+(\partial(\Sigma_1\#\Sigma_2))\oplus\Omega^0_+(\partial\Sigma_3)$, respectively. So (36) and (37) have the same rows.

From the definition, we also have

$$\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}+(\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3}-\bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3})=\bar{\pi}_{\Sigma_1,\Sigma_2}\oplus\pi_{\Sigma_3}$$

and

$$\pi_{\Sigma_1\#\Sigma_2}\oplus\pi_{\Sigma_3}+(\pi_{\Sigma_1\#\Sigma_2}\oplus\pi_{\Sigma_3}-\pi_{(\Sigma_1\#\Sigma_2)\sqcup\Sigma_3})=\pi_{\Sigma_1\#\Sigma_2}\oplus\pi_{\Sigma_3}.$$

Similar to the proof of Lemma 1.4.3, we can show that

$$\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3}-\bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3}$$

and

$$\pi_{\Sigma_1\#\Sigma_2}\oplus\pi_{\Sigma_3}-\pi_{(\Sigma_1\#\Sigma_2)\sqcup\Sigma_3}$$

are trace-class operators. Thus by Proposition 1.2.12, the diagram

$$\begin{array}{ccc}
\mathrm{Det}(\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}+(\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3}-\bar{\pi}_{\Sigma_1\sqcup\Sigma_2,\Sigma_3})) & \longrightarrow & \mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} \\
\downarrow & & \downarrow \\
\mathrm{Det}(\pi_{\Sigma_1\#\Sigma_2}\oplus\pi_{\Sigma_3}) & \longrightarrow & \mathrm{Det}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}
\end{array}$$

(38)

is commutative. Similarly, we can show that the diagrams

$$\begin{array}{ccc}
\mathrm{Det}(\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}+(\pi_{\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3}-\bar{\pi}_{\Sigma_1,\Sigma_2\sqcup\Sigma_3})) & \longrightarrow & \mathrm{Det}(\pi_{\Sigma_1}\oplus\pi_{\Sigma_2\#\Sigma_3}) \\
\downarrow & & \downarrow \\
\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} & \longrightarrow & \mathrm{Det}\bar{\pi}_{\Sigma_1,(\Sigma_2\#\Sigma_3)}
\end{array}$$

(39)

is commutative.

By definition and the surjectivity of $\Delta_{\Sigma_1,\Sigma_2\sqcup\Sigma_3}$ and $\Delta_{\Sigma_1\#\Sigma_2,\Sigma_3}$, the diagrams

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Hol}((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3) & \longrightarrow & \mathrm{Hol}(\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3) & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2\sqcup\Sigma_3}} & \\
 & & \downarrow{\scriptstyle\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}} & & \downarrow{\scriptstyle\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}} & & \\
0 & \longrightarrow & \Omega^0_+(\partial((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3)) & \longrightarrow & \Omega^0_+(\partial((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3))\oplus\Omega^0(S_{12}) & \longrightarrow & \\
 & & & \xrightarrow{\Delta_{\Sigma_1,\Sigma_2\sqcup\Sigma_3}} & \Omega^0(S_{12}) & \longrightarrow & 0 \\
 & & & & \downarrow{\scriptstyle I_{\Omega^0(S_{12})}} & & \\
 & & & \longrightarrow & \Omega^0(S_{12}) & \longrightarrow & 0
\end{array}$$

and

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hol}(\Sigma_1\#\Sigma_2\#\Sigma_3) \arrow[r] \arrow[d, "\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3}"'] & \mathrm{Hol}((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3) \arrow[r, "\Delta_{\Sigma_1\#\Sigma_2,\Sigma_3}"] \arrow[d, "\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}"'] & {} \\
0 \arrow[r] & \Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)) \arrow[r] & \Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3))\oplus\Omega^0(S_{23}) \arrow[r] & {}
\end{tikzcd}$$

$$\begin{tikzcd}
{} \arrow[r, "\Delta_{\Sigma_1\#\Sigma_2,\Sigma_3}"] & \Omega^0(S_{23}) \arrow[r] \arrow[d, "I_{\Omega^0(S_{23})}"'] & 0 \\
{} \arrow[r] & \Omega^0(S_{23}) \arrow[r] & 0
\end{tikzcd}$$

are commutative and have exact rows. From these two diagrams, we see that if we embed $\mathrm{Hol}(\Sigma_1\#\Sigma_2\#\Sigma_3)$ into $\mathrm{Hol}((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3)$, $\mathrm{Hol}((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3)$ into $\mathrm{Hol}(\Sigma_1\sqcup\Sigma_2\sqcup\Sigma_3)$, and identify $\Omega^0_+(\partial((\Sigma_1\#\Sigma_2)\sqcup\Sigma_3))$ with

$$\Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3))\oplus\Omega^0(S_{12}),$$

then

$$\begin{aligned}
\mathrm{Ker}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3} &= \mathrm{Ker}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3} \\
&= \mathrm{Ker}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}, && (40)\\
\mathrm{Im}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3} &= \mathrm{Im}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}\cap\Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)), && (41)\\
\mathrm{Im}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3} &= \mathrm{Im}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}\cap(\Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3))\oplus\Omega^0(S_{23})). && (42)
\end{aligned}$$

From (40) and (41), we obtain

$$\mathrm{Im}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3} = \mathrm{Im}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}\cap\Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3)). \tag{43}$$

The canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}$ to $\mathrm{Det}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}$ is induced from the second equality in (40) and equality (42). The canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}$ to $\mathrm{Det}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3}$ is induced from the first equality in (40) and from (41). So the composition of these two canonical isomorphism is induced from the outer equality of (40) and the equality (43).

On the other hand, denoting $\Sigma_1\#\Sigma_2\#\Sigma_3$ by Σ_{123}, we see that the diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hol}(\Sigma_{123}) \arrow[r] \arrow[d, "\pi_{\Sigma_{123}}"'] & \mathrm{Hol}(\Sigma_1 \sqcup \Sigma_2 \sqcup \Sigma_3) \arrow[r, "\Delta_{\Sigma_1,\Sigma_2}\oplus\Delta_{\Sigma_2,\Sigma_3}"] \arrow[d, "\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}"'] & {} \\
0 \arrow[r] & \Omega^0_+(\partial(\Sigma_{123})) \arrow[r] & \Omega^0_+(\partial(\Sigma_{123})) \oplus \Omega^0(S_{12}) \oplus \Omega^0(S_{23}) \arrow[r, "\Delta_{\Sigma_1,\Sigma_2}\oplus\Delta_{\Sigma_2,\Sigma_3}"] \arrow[d, "I_{\Omega^0(S_{12})\oplus\Omega^0(S_{23})}"'] & \Omega^0(S_{12}) \oplus \Omega^0(S_{23}) \arrow[r] & 0 \\
{} \arrow[r] & {} \arrow[r] & \Omega^0(S_{12}) \oplus \Omega^0(S_{23}) \arrow[r] & 0
\end{tikzcd} \tag{44}$$

is commutative and has exact rows. From this diagram, we obtain

$$\begin{aligned}
\mathrm{Ker}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3} &= \mathrm{Ker}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}, \\
\mathrm{Im}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3} &= \mathrm{Im}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} \cap \Omega^0_+(\partial(\Sigma_1\#\Sigma_2\#\Sigma_3))
\end{aligned}$$

which are exactly the outer equality of (40) and the equality (43), respectively. So we have shown that the composition of the canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}$ to $\mathrm{Det}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}$ and the canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3}$ to $\mathrm{Det}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3}$ is equal to the canonical isomorphism obtained from the diagram (44).

Similarly, we can show that the composition of the canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3}$ to $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2\#\Sigma_3}$ and the canonical isomorphism from $\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2\#\Sigma_3}$ to $\mathrm{Det}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3}$ is equal to the canonical isomorphism obtained from the diagram (44). Thus the diagram

$$\begin{tikzcd}
\mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2,\Sigma_3} \arrow[r] \arrow[d] & \mathrm{Det}\bar{\pi}_{\Sigma_1,\Sigma_2\#\Sigma_3} \arrow[d] \\
\mathrm{Det}\bar{\pi}_{\Sigma_1\#\Sigma_2,\Sigma_3} \arrow[r] & \mathrm{Det}\pi_{\Sigma_1\#\Sigma_2\#\Sigma_3}
\end{tikzcd} \tag{45}$$

is commutative. Putting the commutative diagrams (35), (38), (39) and (45) together, we obtain the diagram (33), and consequently it is commutative. □

For $g, m, n \in \mathbb{N}$, let $\mathcal{B}_{g,m,n}$ be the moduli space of all genus-g connected compact Riemann surfaces with m positively oriented and n negatively oriented analytically parametrized boundary components. Now we would like to give a holomorphic line bundle structure to the disjoint union of determinant lines over elements of $\mathcal{B}_{0,m,n}$ for any $m \in \mathbb{Z}_+$ and $n \in \mathbb{N}$.

For each analytic isomorphism class of Riemann surfaces with parametrized boundaries, we select one Riemann surface Σ with parametrized

boundary in this class as a representative, and we define the determinant line of the isomorphism class $[\Sigma]$ to be the determinant line of Σ. We denote this determinant line by $\mathrm{Det}_{[\Sigma]}$. Let $m \in \mathbb{Z}_+$ and $n \in \mathbb{N}$. We would like to give a holomorphic line bundle structure to $\sqcup_{[\Sigma]\in\mathcal{B}_{0,m,n}}\mathrm{Det}_{[\Sigma]}$. First we prove that $\mathrm{Det}_{[\Sigma]}$ is $\mathbb{C}$, so that the choice of representatives is immaterial and in fact we show:

LEMMA 1.4.5. *For any genus-zero Riemann surface with parametrized boundary* Σ, $\mathrm{Ker}\pi_\Sigma = 0$ *if* Σ *has at least one positively oriented boundary component,* $\mathrm{Ker}\pi_\Sigma = \mathbb{C}$ *if* Σ *has no positively oriented boundary component, and* $\mathrm{Coker}\pi_\Sigma = 0$ *in either case. In particular,* $\mathrm{Det}_{[\Sigma]}$ *is* $\mathbb{C}$ *and the degree of* $\mathrm{Det}_\Sigma = \mathbb{C}$ *is* 0 *if* Σ *has at least one positively oriented boundary component and is* 1 *if* Σ *has no positively oriented boundary component.*

PROOF. The kernel is already calculated in Corollary 1.3.2. We only calculate the cokernel here. Any genus-zero Riemann surface with parametrized boundary is conformally equivalent to a connected region in $\mathbb{C} \subset \widehat{\mathbb{C}}$ with analytically parametrized boundary components such that the parametrization of any boundary component can be extended to an analytic diffeomorphism from the closed unit disk to the union of the boundary component and its interior (exterior) in $\widehat{\mathbb{C}}$ if the interior (exterior) is not in the region. Thus we can assume that Σ is such a genus-zero Riemann surface with parametrized boundary.

Assume that the boundary components of Σ are C_i, $i = 1, \dots, k$, and the interiors of C_i, $i = 1, \dots, k-1$, and the exterior of C_k are not in Σ. If $g \in \Omega^0_+(\partial\Sigma)$, we define a function f on Σ by

$$f(z) = \frac{1}{2\pi i}\int_{\partial\Sigma}\frac{g(\xi)}{\xi - z}d\xi$$

for $z \in \Sigma$. Using the same method as in the proof of Lemma 1.4.1, it can be shown easily that $f|_{\partial\Sigma} = g$ and thus $\pi_\Sigma(f) = g$. So π_Σ is onto and consequently $\mathrm{Coker}\pi_\Sigma = 0$. □

An immediate consequence of this result is the following:

PROPOSITION 1.4.6. *For any* $m \in \mathbb{Z}_+$, $n \in \mathbb{N}$, $\sqcup_{[\Sigma]\in\mathcal{B}_{0,m,n}}\mathrm{Det}_{[\Sigma]}$ *is canonically in bijection with* $\mathcal{B}_{0,m,n} \times \mathbb{C}$. ∎

Now we have:

PROPOSITION 1.4.7. *For any* $m \in \mathbb{Z}_+$, $n \in \mathbb{N}$, $\mathcal{B}_{0,m,n}$ *is an infinite-dimensional complex Banach manifold and* $\sqcup_{[\Sigma]\in\mathcal{B}_{0,m,n}}\mathrm{Det}_{[\Sigma]}$ *forms a trivial holomorphic line bundle over* $\mathcal{B}_{0,m,n}$.

PROOF. In the case on $m = 1$, the moduli space $\mathcal{B}_{0,1,n}$ is the same as the moduli space $K_{\mathfrak{H}_1}(n)$ which we know is an infinite-dimensional complex Banach manifold (see Section 6.4). For $m \neq 1$, it can be shown similarly that $\mathcal{B}_{0,m,n}$ is an infinite-dimensional complex Banach manifold. By Proposition 1.4.6, we see that $\sqcup_{[\Sigma]\in\mathcal{B}_{0,m,n}}\mathrm{Det}_{[\Sigma]}$ forms a trivial holomorphic line bundle over $\mathcal{B}_{0,m,n}$. □

The holomorphic line bundle over $\mathcal{B}_{0,m,n}$ obtained in Proposition 1.4.7 is called the *determinant line bundle over* $\mathcal{B}_{0,m,n}$ and the trivialization that we have just used to define the holomorphic line bundle structure is called the *standard trivialization*.

For determinant line bundles over finite-dimensional complex manifolds, see for example [**?**] [**?**], [**?**], and [**?**].

From now on we shall discuss only determinant lines over genus-zero Riemann surfaces with parametrized boundaries. It is clear that the canonical isomorphisms of determinant lines constructed above induce isomorphisms of line bundles. These isomorphisms between line bundles are still called the *canonical isomorphisms associated to sewing*. We have the following stronger result:

PROPOSITION 1.4.8. *The canonical isomorphisms associated to sewing are holomorphic.*

PROOF. Note that a canonical isomorphisms associated to sewing is constructed in two steps: The first step is the construction using the snake lemma (Proposition 1.2.3) of the canonical isomorphism between the determinant lines of $\pi_{\Sigma_1\#\Sigma_2}$ and $\bar{\pi}_{\Sigma_1,\Sigma_2}$. The second step is the construction using Proposition 1.2.10 of the canonical isomorphism between the determinant lines of $\bar{\pi}_{\Sigma_1,\Sigma_2}$ and $\pi_{\Sigma_1\sqcup\Sigma_2}$.

From the proofs of Propositions 1.2.3 and 1.2.10, we see that if $\pi_{\Sigma_1\#\Sigma_2}$, $\bar{\pi}_{\Sigma_1,\Sigma_2}$ and $\pi_{\Sigma_1\sqcup\Sigma_2}$ depend on some complex parameters holomorphically, the canonical isomorphisms constructed in both steps also depend on these parameters holomorphically. Thus we need only show that $\pi_{\Sigma_1\#\Sigma_2}$, $\bar{\pi}_{\Sigma_1,\Sigma_2}$ and $\pi_{\Sigma_1\sqcup\Sigma_2}$ depend on the elements of the moduli spaces holomorphically. Since these maps are obtained from restrictions and decompositions of subspaces of Sobolev spaces, we need only show that these subspaces, restrictions and decompositions depend holomorphically on the elements of the moduli spaces.

We now prove the holomorphicity of these structures. We show first that the spaces of holomorphic functions on genus-zero Riemann surfaces with parametrized boundaries form holomorphic bundle over the moduli spaces of genus-zero Riemann surfaces with parametrized boundaries.

We only discuss the moduli space of genus-zero Riemann surfaces with one positively oriented boundary component and n negatively oriented boundary components where $n \in \mathbb{Z}_+$. The other cases are the same. By Section 6.4, the moduli space of genus-zero Riemann surfaces with one positively oriented boundary component and n negatively oriented boundary components is a submanifold of the moduli space of spheres with tubes of type $(1, n)$. Using the discussion in Section 1.3, we can identify this moduli space with the space of $n+1$ analytic invertible maps $g_0, \dots, g_n$ defined on the closed unit disk such that the images of these maps are disjoint from each other and $g_0(0) = \infty$, $(1/g_0(w))'|_{w=0} = 1$ and $g_n(0) = 0$. (Note that $g_i(0)$, $i = 1, \dots, n-1$, are the punctures not equal to ∞ and 0.) Then a canonical representative Σ of an element of the moduli space is given by deleting the images of the open unit disk under g_i, $i = 0, \dots, n$ and taking the boundary parametrizations to be the restrictions of g_i, $i = 0, \dots, n$, to the unit circle S^1. We would like to give a holomorphic bundle structure to the disjoint union of the spaces $\mathrm{Hol}(\Sigma)$ for all such Σ.

Let $S_i = g_i(S^1)$. Using the Cauchy formula, we see that for any $h \in \mathrm{Hol}(\Sigma)$,

$$h(z) = \sum_{i=0}^{n} \int_{S_i} \frac{h(\zeta)}{\zeta - z} d\zeta$$

for $z \in \Sigma$. For any i, $\int_{S_i} \frac{h(\zeta)}{\zeta - z} d\zeta$ gives a holomorphic function outside and on S_i. Thus we need only give a holomorphic bundle structure to functions holomorphic outside and on S_i for any fixed i.

We discuss the case $i = n$. To construct a holomorphic bundle structure on the space of functions holomorphic outside and on S_n, we construct a bijection between this space and the space of functions holomorphic outside and on S^1 as follows: Let f be a function holomorphic outside and on S_n and $f|_{S_n}$ the restriction of f to S_n. The restriction $g_n|_{S^1}$ of g_n to S^1 pulls f (or equivalently $f|_{S_n}$) back to a smooth function $f \circ (g_n|_{S^1})$ on S^1. Taking the part of $f \circ (g_n|_{S^1})$ which can be holomorphically extended outside S^1 to be the image of f, we obtain a linear map. We can also construct a map from the space of functions holomorphic outside and on S^1 to the space of functions holomorphic outside and on S_n using the same method together with Lemma 1.4.2. Since g_n is a holomorphic map from the closed unit disk to the union of S_n and the interior of S_n, it is easy to see that these two maps are inverse to each other and therefore are indeed bijections. Using these bijections, we obtain a holomorphic bundle structure. From

the definition of this holomorphic structure, it is clear that the restriction operation taking holomorphic functions on the surface to functions on the boundary components are holomorphic.

The discussion above implies that $\overline{\mathrm{Hol}}_{(s)}(\Sigma_1\#\Sigma_2)$ and $\overline{\mathrm{Hol}}_{(s)}(\Sigma_1 \sqcup \Sigma_2)$ form holomorphic bundles over the moduli spaces. The spaces $H_{(s-\frac{1}{2})}(\partial(\Sigma_1\#\Sigma_2))$ and $H_{(s-\frac{1}{2})}(\partial(\Sigma_1 \sqcup \Sigma_2))$ clearly form holomorphic bundles over the moduli spaces. Thus the only thing that we still need to show is the holomorphicity of the decompositions of $H_{(s-\frac{1}{2})}(\partial(\Sigma_1\#\Sigma_2))$ and $H_{(s-\frac{1}{2})}(\partial(\Sigma_1 \sqcup \Sigma_2))$. But this holomorphicity is obvious since these decompositions are all defined using the decomposition of functions on S^1.

Another proof of this holomorphicity suggested by the proof above has been supplied by Deligne: We still use the description of the moduli space of genus-zero Riemann surfaces with one positively oriented boundary component and n negatively oriented boundary components in terms of g_i, $i = 0, \dots, n$ as above. Note that the restriction of a function in $\mathrm{Hol}(\Sigma)$ to $\partial\Sigma$ gives an embedding of $\mathrm{Hol}(\Sigma)$ into $\Omega^0(\partial\Sigma)$. Using g_i, $i = 0, \dots, n$, we obtain an isomorphism between $\Omega^0(\partial\Sigma)$ and $\oplus_{i=0}^n \Omega^0(S^1)$. Thus $\mathrm{Hol}(\Sigma)$ is also embedded in $\oplus_{i=0}^n \Omega^0(S^1)$.

We now view $\mathrm{Hol}(\Sigma)$ as a subspace of $\oplus_{i=0}^n \Omega^0(S^1)$. We observe that the holomorphicity we would like to prove is equivalent to the holomorphicity of the dependence of $\mathrm{Hol}(\Sigma) \subset \oplus_{i=0}^n \Omega^0(S^1)$ on g_i, $i = 0, \dots, n$. For any $f \in \Omega^0(\partial\Sigma)$, by Lemma 1.4.2, the limit of $\int_{\partial\Sigma} \frac{f(\zeta)}{\zeta - z} d\zeta$ when z goes from the interior of Σ to a point on $\partial\Sigma$ exists and these limits give an element of $\Omega^0(\partial\Sigma)$. Thus we obtain an endomorphism of $\Omega^0(\partial\Sigma)$. The isomorphism from $\Omega^0(\partial\Sigma)$ to $\oplus_{i=0}^n \Omega^0(S^1)$ push this endomorphism forward to an endomorphism of $\oplus_{i=0}^n \Omega^0(S^1)$. This endomorphism of $\oplus_{i=0}^n \Omega^0(S^1)$ depends holomorphically on g_i, $i = 0, \dots, n$ because it is given by the limit of a sum of integrals whose integrands are $f(g_i)dg_i/(g_i - z)$, $i = 0, \dots, n$, depending holomorphically on g_i. Note that this endomorphism of $\oplus_{i=0}^n \Omega^0(S^1)$ is the identity on functions in $\mathrm{Hol}(\Sigma) \in \Omega^0(\partial\Sigma)$, and is 0 on functions in $\Omega^0(\partial\Sigma)$ which can be extended holomorphically to the images of the closed unit disk under g_i, $i = 0, \dots, n$. So $\mathrm{Hol}(\Sigma)$ is in fact the 1-eigenspace of an idempotent varying holomorphically with g_i, $i = 0, \dots, n$. Thus $\mathrm{Hol}(\Sigma)$ depends holomorphically on g_i, $i = 0, \dots, n$. $\square$

Now we can easily construct holomorphic flat connections on the determinant line bundles over $\mathcal{B}_{0,m,n}$ for $m \in \mathbb{Z}_+, n \in \mathbb{N}$. By Proposition 1.4.7, the determinant line bundle over $\mathcal{B}_{0,m,n}$ is trivial for any $m, n \in \mathbb{N}$ not both 0. Using the standard trivialization, we have an

isomorphism from the determinant line bundle over $\mathcal{B}_{0,m,n}$ to the product bundle $\mathcal{B}_{0,m,n} \times \mathbb{C}$. The pullback of the trivial connection on the product bundle under this isomorphism gives a holomorphic flat connection on the determinant line bundle over $\mathcal{B}_{0,m,n}$. This connection is called the *canonical connection*.

REMARK 1.4.9. *Assume that we have a connection on $\mathcal{B}_{0,m,n}$ for any $m, n \in \mathbb{N}$ not both 0. Let $[\Sigma_1] \in \mathcal{B}_{0,m,n}$, $[\Sigma_2] \in \mathcal{B}_{0,k,l}$, $[\Sigma] = [\Sigma_1 \# \Sigma_2]$, $\lambda_1 \in \mathrm{Det}_{\Sigma_1}$, $\lambda_2 \in \mathrm{Det}_{\Sigma_2}$ and $\lambda = \ell_{\Sigma_1,\Sigma_2}(\lambda_1 \otimes \lambda_2)$. The connections give a horizontal subspace $H_{\Sigma,\lambda}$ of the tangent space at $([\Sigma], \lambda)$ of the determinant line bundle over $\mathcal{B}_{0,m+k-1,n+l-1}$ and horizontal subspaces H_{Σ_1,λ_1} and H_{Σ_2,λ_2} of the tangent spaces at $([\Sigma_1], \lambda_1)$ and at $([\Sigma_2], \lambda_2)$ of the determinant line bundles over $\mathcal{B}_{0,m,1}$ and $\mathcal{B}_{0,1,n}$, respectively. If for all $[\Sigma_1] \in \mathcal{B}_{0,m,n}$, $[\Sigma_2] \in \mathcal{B}_{0,k,l}$, $\lambda_1 \in \mathrm{Det}_{\Sigma_1}$ and $\lambda_2 \in \mathrm{Det}_{\Sigma_2}$, $H_{\Sigma,\lambda}$ is equal to the push-forward of $H_{\Sigma_1,\lambda_1} \oplus H_{\Sigma_2,\lambda_2}$ under the sewing map*

$$\begin{aligned} \# : \mathcal{B}_{0,m,n} \times \mathcal{B}_{0,k,l} &\to \mathcal{B}_{0,m+k-1,n+l-1}, \\ (\Sigma_1, \Sigma_2) &\mapsto \Sigma_1 \# \Sigma_2, \end{aligned}$$

then we say that these connections are compatible with the sewing. Note that the canonical connection is not compatible with sewing because the canonical isomorphism from $\mathrm{Det}_{\Sigma_1} \otimes \mathrm{Det}_{\Sigma_2} = \mathbb{C} \otimes \mathbb{C}$ to $\mathrm{Det}_{\Sigma} = \mathbb{C}$ is in general not the trivial isomorphism. But if we consider the subspace of $\mathcal{B}_{0,m,n}$ consisting of conformal equivalence classes of copies of the standard sphere $\widehat{\mathbb{C}}$ with $m+n$ disjoint disks deleted and with dilations and rotations as parametrizations of the boundary components, it is easy to see from the construction of the canonical isomorphisms that the restrictions of the canonical connections to this subspace are in fact compatible with the sewing.

1.5. The determinant line bundle and central extensions of the diffeomorphism group of S^1

Let $\mathrm{Diff}^+(S^1)$ be the group of orientation preserving diffeomorphism of S^1. In this section, we show that the determinant line bundle constructed in Section D.3 gives a central extension of $\mathrm{Diff}^+(S^1)$ of central charge 2.

We first need to construct a central extension of $\mathrm{Diff}^+(S^1)$ using restricted general linear groups (or Shale groups) first studied by Shale [**?**]. Let H be a Hilbert space with a polarization, that is, a decomposition $H = H_+ \oplus H_-$ as the direct sum of two closed orthogonal subspaces H_+ and H_-. Then any linear bounded operator on H can

be written as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{46}$$

where the matrix elements $a : H_+ \to H_+$, $b : H_- \to H_+$, $c : H_+ \to H_-$ and $d : H_- \to H_-$ are bounded linear operators. From now on we shall follow [**?**] to use $A, A_1, \dots$ to denote bounded linear operators on H and $a, b, c, d, a_1, b_1, c_1, d_1, \dots$ to denote the corresponding matrix elements.

The *restricted general linear group* (or *Shale group* $GL_{\text{res}}(H)$ is the set of all invertible bounded linear operators A on H such that b and c are Hilbert-Schmidt operators. Thus for any element $A \in GL_{\text{res}}(H)$, a and d must be Fredholm operators since the existence of A^{-1} and the compactness of b and c imply that a and d have quasi-inverses.

Let $\widetilde{GL}_{\text{res}}(H)$ be the set of all pairs (A, λ) where $A \in GL_{\text{res}}(H)$ and $\lambda \in \text{Det}a$. Let $(A_1, \lambda_1), (A_2, \lambda_2) \in \widetilde{GL}_{\text{res}}(H)$ and $A_3 = A_1A_2$. Then $a_1a_2 + b_1c_2 = a_3$. Since b_1 and c_2 are Hilbert-Schmidt operators, b_1c_2 is of trace class. By Proposition 1.2.11, $\text{Det}(a_1a_2) \simeq \text{Det}a_3$. But by Proposition 1.2.4, $\text{Det}(a_1a_2) \simeq \text{Det}a_1 \otimes \text{Det}a_2$. So $\text{Det}a_1 \otimes \text{Det}a_2$ is canonically isomorphic to $\text{Det}a_3$. Let λ_3 be the image of $\lambda_1 \otimes \lambda_2$ under this canonical isomorphism. We define the product of (A_1, λ_1) and (A_2, λ_2) to be (A_3, λ_3). It is not difficult to verify using the constructions in Section D.2 that this product is associative.

For the identity operator I_H on H, the corresponding matrix element mapping H_+ to itself is the identity I_{H_+}. But $\text{Det}I_{H_+} = \mathbb{C}$. So $1 \in \text{Det}I_{H_+}$. We shall denote $1 \in \text{Det}I_{H_+}$ by ϵ. The element (I_H, ϵ) is an identity for the product defined above. For any $(A, \lambda) \in \widetilde{GL}_{\text{res}}(H)$, let A_0 be the inverse of A. (We use A_0 instead of A^{-1} because we would like to use our convention on the notation above.) Then a_0 is a quasi-inverse of a. So we have $\text{Det}a \otimes \text{Det}a_0 \simeq \text{Det}I_{H_+}$. In particular, we can find a unique element $\mu \in \text{Det}a_0$ such that $\lambda \otimes \mu$ is mapped to ϵ under the canonical isomorphism above from $\text{Det}a \otimes \text{Det}a_0$ to $\text{Det}I_{H_+}$. Then (A_0, μ) is an inverse of (A, λ). We see that $\widetilde{GL}_{\text{res}}(H)$ is a group. We can also give an analytic structure to $\widetilde{GL}_{\text{res}}(H)$ such that it becomes a Lie group. Clearly it is a central extension of $GL_{\text{res}}(H)$ by $\mathbb{C}^\times$.

For any central extension $\tilde{G}$ of a group G by $\mathbb{C}^\times$, a map $\sigma : G \to \tilde{G}$ is called a *section* if the composition of σ and the homomorphism from $\tilde{G}$ to G is the identity of G. Given any section σ, $\sigma(g_1)\sigma(g_2)\sigma(g_1g_2)^{-1}$ is in the center of $\tilde{G}$ and thus can be identified with an element $C(g_1, g_2)$ of $\mathbb{C}^\times$. This function $C : G \times G \to \mathbb{C}$ is called the *cocycle associated to the central extension* $\tilde{G}$ *and the section* σ. If G is a Lie group, $\tilde{G}$ is a

Lie group central extension of G by $\mathbb{C}^\times$ and σ is a smooth section, we can differentiate the cocycle associated to them to obtain a Lie algebra cocycle in the following way: Let $\mathfrak{g}$ be the Lie algebra of G. The *Lie algebra cocycle corresponding to the central extension* $\tilde{G}$ *and the section* σ is the map $c : \mathfrak{g} \otimes \mathfrak{g} \to \mathbb{C}$ defined by

$$c(X,Y) = \frac{\partial^2}{\partial t_1 \partial t_2} C(e^{t_1 X}, e^{t_2 Y})\bigg|_{t_1=t_2=0} - \frac{\partial^2}{\partial t_1 \partial t_2} C(e^{t_2 Y}, e^{t_1 X})\bigg|_{t_1=t_2=0}$$

for $X, Y \in \mathfrak{g}$.

In our case, there is no continuous sections for the central extension $\widetilde{GL}_{\mathrm{res}}(H)$ of $GL_{\mathrm{res}}(H)$. But if we have a smooth section defined on a dense open subset containing the identity of the identity component of GL_{res}, we can still define the cocycle and the Lie algebra cocycle associated to the central extension $\widetilde{GL}_{\mathrm{res}}(H)$ and the section in the same way. The identity component $GL^0_{\mathrm{res}}(H)$ consists of all elements A of $GL_{\mathrm{res}}(H)$ such that the index of a is 0. One dense open subset of $GL^0_{\mathrm{res}}(H)$ is the set U of all $A \in GL_{\mathrm{res}}(H)$ such that a is invertible. For any $A \in U$, $\mathrm{Det}a = \mathbb{C}$. For any $a \in GL(H_+)$, we denote $1 \in \mathbb{C} = \mathrm{Det}a$ by ϵ_a to avoid confusion. Then we have a smooth section σ defined on U by $\sigma(A) = (A, \epsilon_a)$ for $a \in U$. Let $A_1, A_2 \in U$ and assume that $A_3 = A_1 A_2 \in U$. Then $a_3 = a_1 a_2 + b_1 c_2$. Since b_1 and c_2 are Hilbert-Schmidt operators, $b_1 c_2$ is of trace class. Thus $a_1 a_2 a_3^{-1}$ has a determinant and by Corollary 1.2.11, the canonical isomorphism from $\mathrm{Det}a_1 a_2 = \mathbb{C}$ to $\mathrm{Det}a_3 = \mathbb{C}$ maps $\epsilon_{a_1 a_2}$ to $\det(a_1 a_2 a_3^{-1})\epsilon_{a_3}$. On the other hand, by Proposition 1.2.4, we see that the canonical isomorphism from $\mathrm{Det}a_1 \otimes \mathrm{Det}a_2$ to $\mathrm{Det}a_1 a_2$ maps $\epsilon_{a_1} \otimes \epsilon_{a_2}$ to $\epsilon_{a_1 a_2}$. Thus

$$\sigma(A_1)\sigma(A_2) = (A_1 A_2, \det(a_1 a_2 a_3^{-1})\epsilon_{a_3}) = \det(a_1 a_2 a_3^{-1})\sigma(A_3)$$

where we have embedded $\mathbb{C}^\times$ into $\widetilde{GL}_{\mathrm{res}}(H)$. So the cocycle corresponding to this central extension and the section σ is $\det(a_1 a_2 a_3^{-1})$.

It is easy to calculate the corresponding Lie algebra cocycle. Let A_1 and A_2 be two elements of the Lie algebra of U. Then they can be represented as bounded linear operators on H. If we still use A_1 and A_2 to denote these operators and use our convention above, then b_1, c_1, b_2 and c_2 are Hilbert-Schmidt operators. Let $A_3 = [A_1, A_2]$. Then

$$[a_1, a_2] - a_3 = c_1 b_2 - b_1 c_2$$

is a trace-class operator. An easy calculation gives:

PROPOSITION 1.5.1. *The Lie algebra cocycle associated to* $\widetilde{GL}_{\mathrm{res}}(H)$ *and* σ *defined above is given by*

$$c(A_1, A_2) = \mathrm{tr}([a_1, a_2] - a_3). \tag{47}$$

∎

REMARK 1.5.2. *Note that both the cocycle and the Lie algebra cocycle corresponding to the central extension* $\widetilde{GL}_{\mathrm{res}}(H)$ *and the section* σ *are the same as those associated to the central extension* $\widetilde{GL_{\mathrm{res}}}(H)$ *of* $GL_{\mathrm{res}}(H)$ *in* [**?**] *and a section over* U *given in Section 6.6 of* [**?**]. *In fact, it is easy to show that these two central extensions are actually equivalent and under the equivalence, the section* σ *and the one in Section 6.6 of* [**?**] *are the same.*

Now we can use the central extension above to obtain a central extension of $\mathrm{Diff}^+(S^1)$. We first need to embed $\mathrm{Diff}^+(S^1)$ in $GL_{\mathrm{res}}(H_{(s)}(S^1))$ for any $s \in \mathbb{R}_+$. For any $f \in H_{(s)}(S^1)$ and $\phi \in \mathrm{Diff}^+(S^1)$, we define $\phi(f)$ by $\phi(f)(e^{i\theta}) = f(\phi^{-1}(e^{i\theta}))$ for $e^{i\theta} \in S^1$. We have the following result (cf. Proposition (5.3) in [**?**]):

PROPOSITION 1.5.3. *The action of* $\mathrm{Diff}^+(S^1)$ *on* $H_{(s)}(S^1)$ *above gives an embedding of* $\mathrm{Diff}^+(S^1)$ *in* $GL_{\mathrm{res}}(H_{(s)}(S^1))$.

PROOF. The proof is completely analogous to that in [**?**] except that the inner product we use is different. We know that $\{e^{in\theta}/(1+|n|)^s\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $H_{(s)}(S^1)$. Let

$$\phi_{mn} = (e^{im\theta}/(1+|n|)^s, \phi(e^{in\theta}/(1+|n|)^s))_{H_{(s)}(S^1)}, \quad m, n \in \mathbb{Z}.$$

We show below that $\sum_{m\in\mathbb{N}, n\in-\mathbb{Z}_+} |\phi_{mn}| < \infty$ and $\sum_{m\in-\mathbb{Z}_+, n\in\mathbb{N}} |\phi_{mn}| < \infty$. These convergences imply that the action of ϕ is indeed in $GL_{\mathrm{res}}(H_{(s)}(S^1))$.

Let $\tilde{\phi}^{-1}$ be the diffeomorphism from $\mathbb{R}$ to itself covering ϕ^{-1}, that is, a diffeomorphism from $\mathbb{R}$ to itself such that $e^{i\tilde{\phi}^{-1}(\theta)} = \phi^{-1}(\theta)$. Then

$$\phi_{mn} = \frac{1}{2\pi}\int_0^{2\pi} (1+|n|)^{-s} e^{in\tilde{\phi}^{-1}(\theta) - im\theta} d\theta.$$

For $m \in -\mathbb{Z}_+, n \in \mathbb{N}$,

$$\psi^{(mn)} = \frac{n}{n-m}\tilde{\phi}^{-1} + \left(1 - \frac{n}{n-m}\right) I_{\mathbb{R}} = \frac{1}{n-m}(n\tilde{\phi}^{-1} - mI_{\mathbb{R}})$$

is also a diffeomorphism from $\mathbb{R}$ to itself satisfying $\psi^{(mn)}(\theta+2\pi) = \psi(\theta)+2\pi$. In terms of $\psi^{(mn)}$, we can write ϕ_{mn}, $m \in -\mathbb{Z}_+, n \in \mathbb{N}$, as

$$\begin{aligned}\phi_{mn} &= \frac{(1+|n|)^{-s}}{2\pi}\int_0^{2\pi} e^{i(n-m)\psi^{(mn)}(\theta)}d\theta \\ &= \frac{(1+|n|)^{-s}}{2\pi}\int_0^{2\pi} e^{i(n-m)\theta}\frac{d(\psi^{(mn)})^{-1}(\theta)}{d\theta}d\theta \\ &= \cdots \\ &= \frac{(1+|n|)^{-s}i^k(n-m)^{-k}}{2\pi}\int_0^{2\pi} e^{i(n-m)\theta}\frac{d^{k+1}(\psi^{(mn)})^{-1}(\theta)}{d\theta^{k+1}}d\theta\end{aligned} \tag{48}$$

for any $k \in \mathbb{N}$. The function on $[0,2\pi]\times[0,1]$ defined by $(\theta,t) \mapsto t\tilde{\phi}^{-1}(\theta)+(1-t)\theta$ is clearly smooth. Thus the absolute value of its $k+1$-st derivative with respect to θ has a maximum value M_k for any $k \in \mathbb{N}$, and we have

$$\left|\frac{d^{k+1}(\psi^{(mn)})^{-1}(\theta)}{d\theta^{k+1}}\right| \leq M_k. \tag{49}$$

By (48) and (49), we obtain

$$|\phi_{mn}| \leq (1+|n|)^{-s}(n-m)^{-k}M_k \leq (n-m)^{-k}M_k$$

for any $k \in \mathbb{N}$. Thus $\sum_{m\in-\mathbb{Z}_+,n\in\mathbb{N}}|\phi_{mn}|$ is convergent. Similarly, the series $\sum_{m\in\mathbb{N},n\in-\mathbb{Z}_+}|\phi_{mn}|$ is also convergent. □

Using this embedding of $\mathrm{Diff}^+(S^1)$ in $GL_{\mathrm{res}}(H_{(s)}(S^1))$, we obtain a central extension of $\mathrm{Diff}^+(S^1)$. We denote this central extension of $\mathrm{Diff}^+(S^1)$ by $\mathrm{Diff}^{+\sim}(S^1)$ and call it the *basic central extension of* $\mathrm{Diff}^+(S^1)$. The complexified Lie algebra of $\mathrm{Diff}^+(S^1)$ has a basis $\{ie^{in\theta}\frac{d}{d\theta}\}_{n\in\mathbb{Z}}$. Using (47) we obtain the following result (cf. Part (b), Section 7 in [**?**]):

PROPOSITION 1.5.4. *The Lie algebra cocycle associated to the basic central extension* $\mathrm{Diff}^{+\sim}(S^1)$ *of* $\mathrm{Diff}^+(S^1)$ *and the section induced from the section* σ *of* $\widetilde{GL}_{\mathrm{res}}(H_{(s)}(S^1))$ *is given by*

$$c\left(ie^{im\theta}\frac{d}{d\theta}, ie^{in\theta}\frac{d}{d\theta}\right) = \frac{1}{6}(m^3-m)\delta_{m+n,0}, \tag{50}$$

for $m,n \in \mathbb{Z}$.

PROOF. The proof is almost identical to that in Part (b) of Section 7 in [**?**]. We know that $\{e^{in\theta}/(i+|n|)^s\}_{n\in\mathbb{Z}}$ is an orthonormal basis of

$H_{(s)}(S^1)$ and $\{e^{in\theta}/(i+|n|)^s\}_{n\in\mathbb{N}}$ is an orthonormal basis of $H^+_{(s)}(S^1)$. For fixed $m, n \in \mathbb{Z}$, let

$$A_1 = ie^{im\theta}\frac{d}{d\theta},$$
$$A_2 = ie^{in\theta}\frac{d}{d\theta}$$

and

$$\begin{aligned} A_3 &= [A_1, A_2] \\ &= (m-n)ie^{i(m+n)\theta}\frac{d}{d\theta}. \end{aligned}$$

Then for any $k \in \mathbb{N}$,

$$a_1a_2\frac{e^{ik\theta}}{(i+|k|)^s} = \begin{cases} \frac{k(n+k)e^{i(m+n+k)\theta}}{(i+|k|)^s} & n+k \geq 0, m+n+k \geq 0, \\ 0 & \text{otherwise}, \end{cases} \tag{51}$$

$$a_2a_1\frac{e^{ik\theta}}{(i+|k|)^s} = \begin{cases} \frac{k(m+k)e^{i(n+m+k)\theta}}{(i+|k|)^s} & m+k \geq 0, n+m+k \geq 0, \\ 0 & \text{otherwise}, \end{cases} \tag{52}$$

$$a_3\frac{e^{ik\theta}}{(i+|k|)^s} = \begin{cases} \frac{-k(m-n)e^{i(m+n+k)\theta}}{(i+|k|)^s} & m+n+k \geq 0, \\ 0 & \text{otherwise}. \end{cases} \tag{53}$$

By (51)–(53),

$$\begin{aligned} &([a_1, a_2] - a_3)\frac{e^{ik\theta}}{(i+|k|)^s} \\ &= \begin{cases} \frac{k(k+m)e^{i(m+n+k)\theta}}{(i+|k|)^s} & n+k \geq 0, m+k < 0, m+n+k \geq 0, \\ \frac{-k(k+n)e^{i(m+n+k)\theta}}{(i+|k|)^s} & n+k < 0, m+k \geq 0, m+n+k \geq 0, \\ 0 & \text{otherwise}. \end{cases} \end{aligned}$$

Thus

$$\begin{aligned} &\mathrm{tr}([a_1, a_2] - a_3) \\ &= \sum_{k\in\mathbb{N}} \left(\frac{e^{ik\theta}}{(i+|k|)^s}, ([a_1, a_2] - a_3)\frac{e^{ik\theta}}{(i+|k|)^s}\right)_{H^+_{(s)}(S^1)} \\ &= \begin{cases} \sum_{\substack{n+k \geq 0, \\ m+n+k \geq 0,}} \sum_{\substack{m+k < 0 \\ k \in \mathbb{N}}} k(k+m) - \sum_{\substack{n+k < 0, \\ m+n+k \geq 0,}} \sum_{\substack{m+k \geq 0 \\ k \in \mathbb{N}}} k(k+n) & m+ \\ 0 & m+ \end{cases} \end{aligned}$$

$$= \begin{cases} \sum_{k=0}^{n} k(k-n) & -m = n \geq 0, \\ -\sum_{k=0}^{n} k(k-m) & m = -n \geq 0, \\ 0 & m+n \neq 0 \end{cases}$$
$$= \frac{1}{6}(m^3 - m)\delta_{m+n,0},$$

proving (50). □

REMARK 1.5.5. *The cocycle (50) is actually expressed using a basis of the complexification of the Lie algebra of* $\mathrm{Diff}^{+\sim}(S^1)$. *The cocycle expressed using a basis of the Lie algebra of* $\mathrm{Diff}^{+\sim}(S^1)$ *(not the complexification) can be obtained easily from (50) using the linearity of the cocycle.*

For a central extension of $\mathrm{Diff}^+(S^1)$, if there is a complex number c and a smooth section defined on a dense open subset containing the identity such that the Lie algebra cocycle corresponding to this central extension and this section is

$$\frac{c}{12}(m^3 - m)\delta_{m+n,0},$$

we call c the *central charge* of the central extension. From (50) we see that the central extension $\mathrm{Diff}^{+\sim}(S^1)$ has central charge 2. From the definition of Lie algebra cocycle, we obtain immediately:

COROLLARY 1.5.6. *Let V be a module for* $\mathrm{Diff}^{+\sim}(S^1)$ *and*

$$R : \mathrm{Diff}^{+\sim}(S^1) \to \mathrm{End}\, V$$

the corresponding representation. Assume that $R_*(\sigma_*(ie^{in\theta}\frac{d}{d\theta}))$, $n \in \mathbb{Z}$, exists in $\mathrm{End}\, V$. Then V is a module for the Virasoro algebra given by $L_n \mapsto R_*(\sigma_*(ie^{in\theta}\frac{d}{d\theta}))$, $n \in \mathbb{Z}$, $d \mapsto 2$. In particular, the central charge of this representation of the Virasoro algebra is 2. ∎

Now we want to establish the relation between the basic central extension of $\mathrm{Diff}^+(S^1)$ and the determinant line bundle constructed in Section D.2. Let $\mathrm{Diff}^+_{\mathrm{an}}S^1$ be the subgroup of Diff^+S^1 consisting of diffeomorphisms of S^1 which can be extended to an analytic map from an annulus containing S^1 to another such annulus. Then the determinant line bundle constructed in E.2 determines a central extension of $\mathrm{Diff}^+_{\mathrm{an}}S^1$ in the following way: Let $\mathbb{A}$ be any Riemann surface with parametrized boundary whose underlying Riemann surface is isomorphic to an annulus and with one boundary component positively oriented while the other is negatively oriented. For any element

$\phi \in \mathrm{Diff}^+_{\mathrm{an}} S^1$, we obtain a Riemann surface with parametrized boundary by composing ϕ with the parametrization of the positively oriented boundary component and denote it by $\phi\mathbb{A}$.

Fix such an $\mathbb{A}$. We define the *determinant line* over ϕ to be

$$\mathrm{Det}_\phi = \mathrm{Det}_{\phi^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^*.$$

Since the positively oriented boundary component of $\mathbb{A}$ is analytically parametrized, we can find a positive number $r_0 < 1$ such that for any r satisfying $1 > r > r_0$, $\mathbb{A} = \mathbb{A}^+ \# \mathbb{A}^-$ where $\mathbb{A}^+$ contains the positively oriented boundary component of $\mathbb{A}$ and is conformally equivalent (via an analytic extension of the parametrization of the positively oriented boundary component of $\mathbb{A}$) to $\mathbb{A}_r$. Thus $\phi^{-1}\mathbb{A} = (\phi^{-1}\mathbb{A}^+)\#\mathbb{A}^-$ and by Theorem 1.4.4

$$\begin{aligned}
\mathrm{Det}_{\phi^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^* &= \mathrm{Det}_{(\phi^{-1}\mathbb{A}^+)\#\mathbb{A}^-} \otimes (\mathrm{Det}_{\mathbb{A}^+\#\mathbb{A}^-})^* \\
&\simeq \mathrm{Det}_{\phi^{-1}\mathbb{A}^+} \otimes \mathrm{Det}_{\mathbb{A}^-} \otimes (\mathrm{Det}_{\mathbb{A}^+})^* \otimes (\mathrm{Det}_{\mathbb{A}^-})^* \\
&\simeq \mathrm{Det}_{\phi^{-1}\mathbb{A}^+} \otimes (\mathrm{Det}_{\mathbb{A}^+})^*. \qquad (54)
\end{aligned}$$

Since (54) holds for any $\mathbb{A}$ above and for all r sufficiently close to 1, we see that Det_ϕ's defined using different $\mathbb{A}$'s are canonically isomorphic.

Let

$$\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1 = \{(\phi, \lambda) \mid \phi \in \mathrm{Diff}^+_{\mathrm{an}} S^1, \lambda \in \mathrm{Det}_\phi\}.$$

We would like to give a group structure to $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$. Let $(\phi_1, \lambda_1), (\phi_2, \lambda_2) \in \widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$. Then using the canonical isomorphisms between Det_ϕ's defined using different $\mathbb{A}$'s, we have

$$\begin{aligned}
\mathrm{Det}_{\phi_1\circ\phi_2} &= \mathrm{Det}_{(\phi_1\circ\phi_2)^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^* \\
&\simeq \mathrm{Det}_{(\phi_1\circ\phi_2)^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\phi_2^{-1}A})^* \otimes \mathrm{Det}_{\phi_2^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^* \\
&= \mathrm{Det}_{\phi_1^{-1}(\phi_2^{-1}\mathbb{A})} \otimes (\mathrm{Det}_{\phi_2^{-1}\mathbb{A}})^* \otimes \mathrm{Det}_{\phi_2^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^* \\
&\simeq \mathrm{Det}_{\phi_1} \otimes \mathrm{Det}_{\phi_2}. \qquad (55)
\end{aligned}$$

Let $\lambda \in \mathrm{Det}_{\phi_1\circ\phi_2}$ be the image of $\lambda_1 \otimes \lambda_2$ under the canonical isomorphism from $\mathrm{Det}_{\phi_1} \otimes \mathrm{Det}_{\phi_2}$ to $\mathrm{Det}_{\phi_1\circ\phi_2}$. We define the product of (ϕ_1, λ_1) and (ϕ_2, λ_2) to be $(\phi_1 \circ \phi_2, \lambda)$. It is clear from (55) that this product is associative. Let I_{S_1} be the identity of $\mathrm{Diff}^+_{\mathrm{an}} S^1$. Then by definition

$$\begin{aligned}
\mathrm{Det}_{I_{S^1}} &= \mathrm{Det}_{I_{S^1}^{-1}\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^* \\
&= \mathrm{Det}_{\mathbb{A}} \otimes (\mathrm{Det}_{\mathbb{A}})^* \\
&\simeq \mathbb{C}. \qquad (56)
\end{aligned}$$

Let ϵ be the preimage of 1 under the canonical isomorphism from $\mathrm{Det}_{I_{S^1}}$ to $\mathbb{C}$ given by (56). Then (I_{S^1}, ϵ) is the identity of $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$. For any

$(\phi, \lambda) \in \widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$, the inverse of (ϕ, λ) is (ϕ^{-1}, μ) where μ is the unique element of $\mathrm{Det}_{\phi^{-1}}$ such that the image of $\lambda \otimes \mu$ under the canonical isomorphism from $\mathrm{Det}_\phi \otimes \mathrm{Det}_{\phi^{-1}}$ to $\mathrm{Det}_{I_{S^1}}$ given by (55) is ϵ. Thus $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$ is a group. We have a natural injective homomorphism $\mathbb{C}^\times \to \widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$ given by $a \mapsto (I_{S^1}, a\epsilon)$ for $a \in \mathbb{C}^\times$. Also we have a natural surjective homomorphism $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1 \to \mathrm{Diff}^+_{\mathrm{an}} S^1$ given by $(\phi, \lambda) \mapsto \phi$ for $(\phi, \lambda) \in \widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$. By definition these maps are homomorphisms. It is easy to see that we can give $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$ an analytic structure so that it becomes an infinite-dimensional Lie group, and such that

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1 \longrightarrow \mathrm{Diff}^+_{\mathrm{an}} S^1 \longrightarrow 1$$

is an exact sequence of Lie groups. We have proved:

PROPOSITION 1.5.7. *The Lie group* $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$ *is a central extension of* $\mathrm{Diff}^+_{\mathrm{an}} S^1$. ∎

The basic central extension $\mathrm{Diff}^{+\sim}(S^1)$ of $\mathrm{Diff}^+(S^1)$ also gives a central extension $\mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$ of $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$. We have:

PROPOSITION 1.5.8. *The central extensions* $\mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$ *and* $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}} S^1$ *are isomorphic.*

PROOF. For any $\phi \in \mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$, let $T_\phi : H^+_{(s-\frac{1}{2})}(S^1) \to H^+_{(s-\frac{1}{2})}(S^1)$ be the operator induced from the action of ϕ on $H_{(s-\frac{1}{2})}(S^1)$. Thus by definition, an element of $\mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$ is a pair (ϕ, λ) where $\phi \in \mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$ and $\lambda \in \mathrm{Det} T_\phi$. Let r be a positive number less than 1. Let

$$C_r^- = \{z \in \mathbb{C} \mid |z| = r\}$$

and

$$C_1^+ = \{z \in \mathbb{C} \mid |z| = 1\}.$$

Then ϕ acts on $H^+_{(s-\frac{1}{2})}(C_1^+)$ by $(\phi(f))(e^{i\theta}) = f(\phi^{-1}(e^{i\theta}))$ for $f \in H^+_{(s-\frac{1}{2})}(C_1^+)$. The operator

$$\pi_{\mathbb{A}_r} : \overline{\mathrm{Hol}}(\mathbb{A}_r) \to H^+_{(s-\frac{1}{2})}(\partial \mathbb{A}_r) = H^+_{(s-\frac{1}{2})}(C_1^+) \oplus H^-_{(s-\frac{1}{2})}(C_r^-)$$

is given by

$$\pi_{\mathbb{A}_r}(f) = ((f|_{C_1^+})_+, (f|_{C_r^-})_-)$$

for $f \in \overline{\mathrm{Hol}}(\mathbb{A}_r)$, where for any $g \in H^+_{(s-\frac{1}{2})}(C_1^+)$ or $g \in H^-_{(s-\frac{1}{2})}(C_r^-)$, g_+ and g_- are the parts of g whose Fourier expansions involve only

nonnegative and negative, respectively, powers of $e^{i\theta}$ such that $g_+ + g_- = g$. The operator

$$\begin{aligned}\pi_{\phi^{-1}\mathbb{A}_r} : \overline{\mathrm{Hol}}(\phi^{-1}\mathbb{A}_r) &= \overline{\mathrm{Hol}}(\mathbb{A}_r)\\ &\to H^+_{(s-\frac{1}{2})}(\partial(\phi^{-1}\mathbb{A}_r)) \simeq H^+_{(s-\frac{1}{2})}(C_1^+) \oplus H^-_{(s-\frac{1}{2})}(C_r^-)\end{aligned}$$

is given by

$$\pi_{\phi^{-1}\mathbb{A}_r}(f) = ((\phi(f|_{C_1^+}))_+, (f|_{C_r^-})_-).$$

Let

$$P_\phi : H^+_{(s-\frac{1}{2})}(\partial\mathbb{A}_r) \to H^+_{(s-\frac{1}{2})}(\partial(\phi^{-1}\mathbb{A}_r))$$

be the linear operator defined as follows: Note that any $f_1 \in H^+_{(s-\frac{1}{2})}(C_1^+)$ and $f_2 \in H^-_{(s-\frac{1}{2})}(C_r^-)$ can be extended to elements of $\overline{\mathrm{Hol}}(\mathbb{A}_r)$. Let $f \in \overline{\mathrm{Hol}}(\mathbb{A}_r)$ be the sum of the extensions of f_1 and f_2. We define P_ϕ by

$$P_\phi((f_1, f_2)) = ((\phi(f|_{C_1^+}))_+, (f|_{C_r^-})_-)$$

for $f_1 \in H^+_{(s-\frac{1}{2})}(C_1^+)$ and $f_2 \in H^-_{(s-\frac{1}{2})}(C_r^-)$. From the definition of P_ϕ, we see that $\pi_{\phi^{-1}\mathbb{A}_r} = P_\phi \circ \pi_{\mathbb{A}_r}$. Thus by Proposition 1.2.4,

$$\mathrm{Det}\pi_{\phi^{-1}\mathbb{A}_r} \simeq \mathrm{Det}P_\phi \otimes \mathrm{Det}\pi_{\mathbb{A}_r}$$

which implies

$$\mathrm{Det}P_\phi \simeq \mathrm{Det}\pi_{\phi^{-1}\mathbb{A}_r} \otimes (\mathrm{Det}\pi_{\mathbb{A}_r})^*. \tag{57}$$

On the other hand, from the definitions of P_ϕ and T_ϕ, we have

$$\begin{aligned}P_\phi((f_1, f_2)) &= ((\phi(f^+|_{C_1^+}))_+ + (\phi(f^-|_{C_1^+}))_+, (f|_{C_r^-})_-)\\ &= (T_\phi(f_1) + (\phi(f^-|_{C_1^+}))_+, f_2),\end{aligned} \tag{58}$$

where f^+ and f^- are the extensions of f_1 and f_2, respectively, to $\overline{\mathrm{Hol}}(\mathbb{A}_r)$.

Let $\overline{T}_\phi$ and Z_ϕ be operators from $H^+_{(s)}(\partial\mathbb{A}_r)$ to $H^+_{(s)}(\partial\phi\mathbb{A}_r)$ defined by

$$\overline{T}_\phi((f_1, f_2)) = (T_\phi(f_1), f_2)$$

and

$$Z_\phi((f_1, f_2)) = ((\phi(f^-|_{C_1^+}))_+, 0),$$

respectively, for $f_1 \in H^+_{(s-\frac{1}{2})}(C_1^+)$ and $f_2 \in H^-_{(s-\frac{1}{2})}(C_r^-)$. Then

$$P_\phi = \overline{T}_\phi + Z_\phi.$$

The map given by $(f_1, f_2) \mapsto (\phi f_-|_{C_1^+})_+$ can be factorized as the composition of the map given by $(f_1, f_2) \mapsto f_2$, the map given by $f_2 \mapsto f^-|_{C_1^+}$

and the map $g \mapsto (\phi(g))_+$. We now show that the map given by $f_2 \mapsto f^-|_{C_1^+}$ is of trace class. In fact, it maps $r^{-n}e^{-in\theta}$ to $e^{-in\theta}$. In general it maps $g \in H^-_{(s-\frac{1}{2})}(C_r^-)$ to

$$\sum_{n\in\mathbb{Z}_+} r^n \left(\frac{e^{-in\theta}}{1+n}, g\right)_{H^-_{(s-\frac{1}{2})}(C_r^-)} \frac{e^{-in\theta}}{1+n}$$

where $(\cdot,\cdot)_{H^-_{(s-\frac{1}{2})}(C_r^-)}$ is the inner product of $H^-_{(s-\frac{1}{2})}(C_r^-)$ defined by

$$(g,h)_{H^-_{(s-\frac{1}{2})}(C_r^-)} = \sum_{n\in\mathbb{Z}_+} (1+n)r^{-2n}\bar{a}_n b_n$$

for $g = \sum_{n\in\mathbb{Z}_+} a_n r^{-n}e^{-in\theta}$ and $h = \sum_{n\in\mathbb{Z}_+} b_n r^{-n}e^{-in\theta}$ in $H^-_{(s-\frac{1}{2})}(C_r^-)$. Since $r < 1$, we see that this map is indeed of trace class. Since the maps given by $(f_1, f_2) \mapsto f_2$ and by $g \mapsto (\phi(g))_+$ are clearly continuous, the map given by $(f_1, f_2) \mapsto (\phi f_-|_{C_1^+})_+$ is of trace class. Thus from its definition, we see that Z_ϕ is also of trace class. Since T_ϕ is Fredholm, we see from the definition of $\overline{T}_\phi$ that $\overline{T}_\phi$ is also Fredholm and $\mathrm{Det}\overline{T}_\phi \simeq \mathrm{Det}T_\phi$. Thus by Proposition 1.2.11, we obtain

$$\mathrm{Det}P_\phi = \mathrm{Det}(\overline{T}_\phi + Z_\phi) \simeq \mathrm{Det}\overline{T}_\phi \simeq \mathrm{Det}T_\phi. \tag{59}$$

Combining (57) and (59), we obtain

$$\mathrm{Det}T_\phi \simeq \mathrm{Det}\pi_{\phi^{-1}\mathbb{A}_r} \otimes (\mathrm{Det}\pi_{\mathbb{A}_r})^*.$$

For any $(\phi,\lambda) \in \mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$, let $\xi((\phi,\lambda)) = (\phi,\mu) \in \widetilde{\mathrm{Diff}}^+_{\mathrm{an}}(S^1)$ where μ is the image of λ under the canonical isomorphism from $\mathrm{Det}T_\phi$ to $\mathrm{Det}\pi_{\phi^{-1}\mathbb{A}_r} \otimes (\mathrm{Det}\pi_{\mathbb{A}_r})^*$. We obtain a map $\xi : \mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1) \to \widetilde{\mathrm{Diff}}^+_{\mathrm{an}}(S^1)$. It is easy to see from the construction that it is indeed an isomorphism. □

For central extensions of $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$, we also have the notion of *central charge* which is defined in the same way as that for central charges of central extensions of $\mathrm{Diff}^+(S^1)$.

Corollary 1.5.9. *The central charge of the central extension* $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}}(S^1)$ *of* $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$ *is* 2.

Proof. By 50, the central charge of the central extension $\mathrm{Diff}^{+\sim}(S^1)$ of $\mathrm{Diff}^+(S^1)$ is 2. Thus the central charge of the central extension $\mathrm{Diff}^{+\sim}_{\mathrm{an}}(S^1)$ of $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$ is also 2. The conclusion follows immediately from Proposition 1.5.8. □

1.6. One-dimensional genus-zero modular functors and the Mumford-Segal theorem

In Section D.3-D.5, we have constructed a determinant line functor from the category of Riemann surfaces with parametrized boundaries to the category of graded lines. This functor is an example of the one-dimensional modular functors introduced by Segal [**?**]. We have constructed the canonical isomorphisms associated with sewing and proved some crucial properties of this functor and the canonical isomorphisms. In addition, this functor has some other properties which we do not discuss since we do not need them for our purposes. For any integer n, the n-th power of the determinant line functor also gives a functor satisfying the same properties.

There is a converse of this construction saying that any functors with these properties must be a power of the functor Det. We call this converse the Mumford-Segal theorem since this theorem appeared first in [**?**] as an analogue of a theorem of Mumford's. This result is discussed further in [**?**] and [**?**]. For simplicity, in this section, we shall discuss only a simple variant of this theorem which is already enough for us here. In fact this section is used only in Section 6.9 and is not needed in the proof of the main theorem of this work.

We restrict our attention in this section to the moduli spaces $\mathcal{B}_{0,1,n}$, $n \in \mathbb{N}$, of genus-zero Riemann surfaces with parametrized boundaries with exactly one positively oriented boundary component. Note that the degree of determinant lines over elements of these moduli spaces are 0 (see Lemma 1.4.5). Thus in this section, we can think of determinant lines simply as one-dimensional vector spaces. The sewing operation $\#$ induces a sewing operation in these moduli spaces. For any $m, n, \in \mathbb{N}$ and any positive integer $i \leq m$, we denote the sewing operation which identifies the i-th negatively oriented boundary component of surfaces in conformal equivalence classes in $\mathcal{B}_{0,1,m}$ with the only positively oriented boundary component of surfaces in conformal equivalence classes in $\mathcal{B}_{0,1,n}$ by ${}_i\#_0$. The sequence $\mathcal{B} = \{\mathcal{B}_{0,1,n}\}_{n\in\mathbb{N}}$ together with the maps ${}_i\#_0$ is an analytic operad.

We now give a variant of Segal's notion of modular functor [**?**] in the genus-zero and one-dimensional case. We shall still use the terminology "one-dimensional genus-zero modular functor" for these objects.

Definition 1.6.1. *A one-dimensional genus-zero modular functor is an analytic operad E together with a morphism $\pi : E \to \mathcal{B}$ of analytic operads and a holomorphic line bundle structure on the triple $(E(n), \mathcal{B}_{0,1,n}, \pi)$ for any $n \in \mathbb{N}$ satisfying the following axiom: Let $[\Sigma_1] \in \mathcal{B}_{0,1,m}$, $[\Sigma_2] \in \mathcal{B}_{0,1,n}$, $m, n \in \mathbb{N}$ and let i be a positive integer*

satisfying $i \leq m$. *Then the map from the Cartesian product of the fibers over* $[\Sigma_1]$ *and* $[\Sigma_2]$ *to the fiber over* $[\Sigma_1]_i\#_0[\Sigma_2]$ *induced from the composition map of* E *is multilinear and gives an isomorphism from the tensor product of the fibers* $[\Sigma_1]$ *and* $[\Sigma_2]$ *to the fiber over* $[\Sigma_1]_i\#_0[\Sigma_2]$.

When there is no confusion, we shall call E a one-dimensional genus-zero modular functor.

EXAMPLE 1.6.2. *The determinant lines over genus-zero Riemann surfaces with parametrized boundaries constructed and discussed in the preceding three sections give an example of one-dimensional genus-zero modular functors. We denote this one-dimensional genus-zero modular functor by* $\mathrm{Det}_{\mathfrak{H}_1}$ *to distinguish it from the vertex partial operad* Det *in Chapter 6. Actually in Chapter 6, we also constructed complex powers of* Det. *These complex powers give complex powers of* $\mathrm{Det}_{\mathfrak{H}_1}$, *which are also examples of one-dimensional genus-zero modular functors. In particular, the* 0*-th power of* $\mathrm{Det}_{\mathfrak{H}_1}$ *is a one-dimensional genus-zero modular functor, which is in fact the trivial one-dimensional genus-zero modular functor.*

Morphisms and *isomorphisms* of one-dimensional genus-zero modular functors are defined in the obvious way.

Given a one-dimensional genus-zero modular functor E, its *dual* E^* of E is a one-dimensional genus-zero modular functor defined in the obvious way.

The following result, which determines all the one-dimensional genus-zero modular functors, is a variant of the Mumford-Segal theorem in the genus-zero case:

THEOREM 1.6.3. *Any one-dimensional genus-zero modular functor is*
isomorphic to a power of $\mathrm{Det}_{\mathfrak{H}_1}$.

PROOF. Similarly to the construction of $\widetilde{\mathrm{Diff}}^+_{\mathrm{an}}(S^1)$, we can construct a central extension of $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$ from a one-dimensional genus-zero modular functor E. From the construction, we see that this central extension depends only on $E(1)$. Let the central charge of this central extension be c. We know that the $c/2$-th power $\mathrm{Det}_{\mathfrak{H}_1}^{c/2}$ of the determinant line bundle also determines a central extension of $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$ of central charge c. Since a central extension of $\mathrm{Diff}^+_{\mathrm{an}}(S^1)$ is uniquely determined up to isomorphism by the Lie algebra cocycle, or equivalently, by the central charge, the central extension obtained from E and from $\mathrm{Det}_{\mathfrak{H}_1}^{c/2}$ must be isomorphic. It is clear that the first component (for

example, $E(1)$ and $\mathrm{Det}_{\mathfrak{H}_1}^{c/2}(1)$) of a one-dimensional genus-zero modular functor together with the composition is a monoid. This structure of a monoid is determined uniquely up to isomorphism by the corresponding central extension of $\mathrm{Diff}_{\mathrm{an}}^{+}(S^1)$. Thus $E(1)$ and $\mathrm{Det}_{\mathfrak{H}_1}^{c/2}(1)$ are isomorphic as monoids.

It is clear that $E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2}$ is also a one-dimensional genus-zero modular functor and $(E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2})(1)$ as a monoid is isomorphic to the first component of the trivial one-dimensional genus-zero modular functor. In particular, the holomorphic line bundle $(E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2})(1)$ is holomorphically trivial and thus has a holomorphic flat connection that is also trivial in the sense that the corresponding representation of the fundamental group $\pi_1(\mathcal{B}_{0,1,1})$ is trivial.

From the discussion in the proof of Proposition 1.4.8, we know that $\mathcal{B}_{0,1,n}$ can be identified with the space of $n+1$ analytic invertible maps $g_0, \dots, g_n$ defined on the closed unit disk such that the images of these maps are disjoint from each other and $g_0(0) = \infty$, $(1/g_0(w))'|_{w=0} = 1$ and $g_n(0) = 0$. Thus $(E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2})(n)$ can be viewed as a holomorphic line bundle over the space of $n+1$-tuples of such maps. Note that any deformation from an element of the moduli space $\mathcal{B}_{0,1,n}$ corresponds to a change of these functions by composing them with suitable analytic functions. But deformations of this type can be obtained by sewing the element we are considering with annulus with one positively oriented and one negatively oriented parametrized boundary components. Thus by the definition of one-dimensional genus-zero modular functor, we see that the holomorphic flat connection on $(E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2})(1)$ gives a holomorphic flat connection on $(E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2})(n)$. This flat connection determines a representation of the fundamental group $\pi_1(\mathcal{B}_{0,1,n})$.

We claim that this representation is trivial. By a theorem of Dehn [**?**] (see also [**?**] or [**?**] or [**?**] for the background material), we know that $\pi_1(\mathcal{B}_{0,1,n})$ is generated by the elements obtained from Dehn twists which can be described in our case as follows: For any $[\Sigma] \in \mathcal{B}_{0,1,n}$ and any decomposition $\Sigma = (\Sigma_1 \, {}_i\#_0 \mathbb{A}_r) {}_i\#_0 \Sigma_2$ where $[\Sigma_1] \in \mathcal{B}_{0,1,m}$, $\Sigma_2 \in \mathcal{B}_{0,1,n}$ and $r > 1$, let $\mathbb{A}_{r,\theta}$ be the same as $\mathbb{A}_r$ except that the parametrization of the negatively oriented boundary components is rotated by the angle θ. Then $\theta \to [(\Sigma_1 \, {}_i\#_0 \mathbb{A}_{r,\theta}) {}_i\#_0 \Sigma_2]$ gives an element of $\pi_1(\mathcal{B}_{0,1,n})$ which we shall call a *Dehn loop at* Σ. To show that the representations of the mapping class groups are trivial, we need only show that these elements act trivially. But this is obvious since Dehn loops are actually in the image of $\mathcal{B}_{0,1,1}$ under the map from $\mathcal{B}_{0,1,1}$ to $\mathcal{B}_{0,1,n}$ given by $[\mathbb{A}] \mapsto [(\Sigma_1 \, {}_i\#_0 \mathbb{A}) {}_i\#_0 \Sigma_2]$ and the connection on $E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2}(1)$ is not just flat

but also trivial. So the representations of the mapping class groups given by the flat connection above must be trivial. This triviality of the representations of the mapping class groups implies that $E^* \otimes \mathrm{Det}_{\mathfrak{H}_1}^{c/2}$ is isomorphic to the trivial one-dimensional genus-zero modular functor and consequently E is isomorphic to $\mathrm{Det}_{\mathfrak{H}_1}^{c/2}$. $\square$